# What Makes a Latent Fire? Discovering Causal Dependencies Between Sparse Autoencoder Latents

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## Abstract

Circuit discovery typically explains an output: a logit, a probability, a behavioural metric. We ask instead what makes an internal latent fire. For a target sparse autoencoder (SAE) latent, *weighted circuit masking* (WCM) learns a circuit of upstream latents with a scaling coefficient for each, trained so that the circuit alone reproduces the target under three different ablations of everything else. At a sparsity penalty that yields circuits of $10^2$–$10^3$ upstream latents, 46% of ∼15,000 sampled latents in TuringLLM, a 12-layer language model, get a circuit that reproduces the target under all three ablations and is necessary for it, on held-out text, while random circuits of the same size, given the same coefficient fitting, fail on all 747 targets tested [*pending: rebuilt held-out re-score, DAN-157; harder random nulls, DAN-181; Pythia-70m and GPT-2 small, DAN-40/41*]. A weaker penalty gives larger circuits that pass more often. The coefficients are what keep these circuits small: without them the same nodes pass for 1.5% of targets instead of 45%, and circuits for targets in the last four layers need $\sim 10^4$ latents to pass, about 10× more, with only half as many of those targets passing at all. The procedure needs no behavioural task and applies to any latent in an SAE bank.

## 1 Introduction

Consider a latent in layer 10 of a 12-layer language model that fires on the possessive in "Ohm's", "Snell's", or "Kirchhoff's", before the word "law" has appeared, but stays silent on "Darwin's", "Rutherford's", and "Picasso's". Given an invented name, it fires in a passage on electromagnetism and not in one on the history of painting. Something upstream supplies what the model knows about which names carry laws, and what the passage is about. What makes this latent fire?

Sparse autoencoders (SAEs) decompose a model's activations into latents like this one, which often track coherent concepts (Cunningham et al., 2024; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024), a working answer to superposition (Elhage et al., 2022). A latent says what the model represents; it does not say how the model computes it. That question belongs to circuit discovery, whose methods, over components or features and by search, attribution, or learned masks, predominantly target an output (Conmy et al., 2023; Syed et al., 2023; Marks et al., 2025; Ameisen et al., 2025; Bhaskar et al., 2024; Caples et al., 2025). An output is the model's computation expressed in our vocabulary; targeting latents instead can capture variables that never show up as a token.

Latent targets have been approached by attribution, tracing upstream of a chosen SAE latent and thresholding the scores (Ge et al., 2024; Golimblevskaia et al., 2025). To our knowledge, no method

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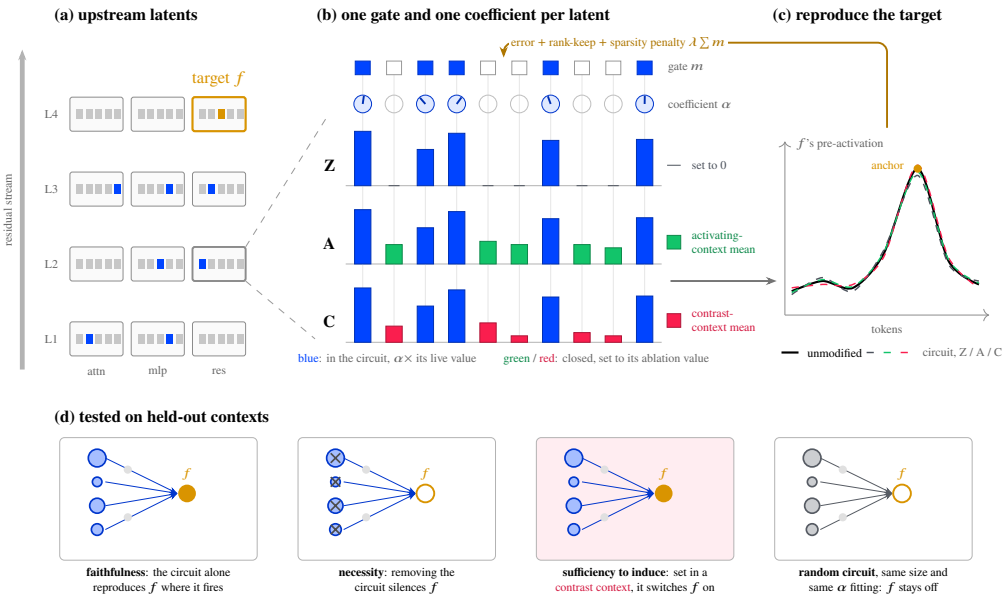



Figure 1: How weighted circuit masking works. **(a)** For a target latent $f$, every latent at every upstream SAE site is a candidate node. **(b)** Each candidate gets a gate $m$ (kept or closed) and a scaling coefficient $\alpha$: a kept latent contributes $\alpha$ times its live value, and a closed one is replaced by its ablation value. **(c)** They are trained so that, under each ablation method, the circuit reproduces $f$'s pre-activation at its anchor positions, with a sparsity penalty on every open gate. **(d)** The weighted circuit, typically $10^2$–$10^3$ latents, is tested on held-out contexts by faithfulness, necessity, and sufficiency to induce, against same-size random circuits with fitted coefficients.

*optimises* a circuit for a single target latent and tests it by intervening on that latent. We call this *latent-targeted circuit discovery* (Figure 1).

Our method, *weighted circuit masking* (WCM), learns two things for every upstream latent: a gate for whether it belongs to the circuit, and a scaling coefficient for how strongly it acts. Both are trained so that the circuit alone reproduces the target with everything else ablated in three ways at once: set to zero, or to its mean over contexts where the target fires, or where it is silent. We test each circuit on held-out text for *faithfulness* (the circuit alone reproduces the target), *necessity* (removing the circuit removes the target), and *sufficiency to induce* (setting the circuit's values switches the target on where it was silent).

We apply WCM to ~15,000 sampled latents of TuringLLM, a 12-layer model trained on educational text [*pending: and to 512 each of Pythia-70m and GPT-2 small, DAN-40/41*]. We fit every target at one sparsity penalty, which yields circuits of $10^2$–$10^3$ upstream latents. At that penalty, 46% of targets get a circuit that passes a strict joint test: faithful under all three ablations and necessary, on held-out text. Random circuits of the same size, given the same coefficient fitting, pass for none of 747 targets [*pending: harder nulls that test the selection itself, DAN-181*]. The rate is one point on a size trade-off, not a ceiling: in a 190-target sweep it rises from 42% to 58% as a weaker penalty grows the median circuit from about 500 to 1,100 latents (Section 5.1 and Appendix H). Residual-stream targets pass most often (65%) and attention outputs least, about one in seven. The coefficients are what keep circuits small: without them, circuits for targets in the last four layers need ~$10^4$ latents to pass, 10× more, and only 41% of those targets pass at any size we tried, against 80% with coefficients (Section 5.2). The study covers Top-K dictionaries only, and node sets are not unique: a refit that passes about as often shares about two thirds of the original circuit's latents, so we report sizes and scores rather than single memberships (Section 6.1 and Appendix O). Section 5.3 opens one passing circuit, for a latent that reads a date or place as classical Greek antiquity; Appendix L returns to the

law latent above, and Appendix M shows circuits for randomly drawn targets.

We contribute:

- **Circuits for arbitrary latents at scale** (Sections 4.2 and 5.1): weighted circuit masking, which learns a gate and a separate scaling coefficient for each upstream latent under three ablations at once, finds circuits of $10^2$–$10^3$ latents that pass a held-out test for 46% of ~15,000 sampled latents, while size-matched random circuits with the same fitting never pass.
- **Coefficients keep circuits small** (Section 5.2): without fitted coefficients the same nodes almost never pass (1.5% against 45%), and the smallest passing circuits for deep targets are about 10× larger, with half as many deep targets passing at all.

Supporting analyses (the production run in detail, specificity, stable cores, the contrast-context source, and the evaluation protocol) are in the appendices, and Section 6 collects recommendations for latent-targeted circuit discovery. We release the pipeline, discovery methods, and analysis code at `github.com/DanielJamesDavies/turing-explorer-circuit-discovery`.

# 2 Related Work

**Circuit discovery.** Manual analysis of components (Olah et al., 2020; Wang et al., 2023) gave way to automated search: ACDC (Conmy et al., 2023) and Edge Pruning (Bhaskar et al., 2024) prune components or edges against an output metric, and attribution patching (Nanda, 2023; Syed et al., 2023; Kramár et al., 2024) and its refinements (Hanna et al., 2024; Rezaei Jafari et al., 2025) score them to first order (our IG baseline adapts EAP-IG's estimator to nodes). Sparse feature circuits (Marks et al., 2025) moved the nodes to SAE features with error nodes, and attribution graphs (Ameisen et al., 2025; Lindsey et al., 2025) build per-prompt graphs over transcoder features (Dunefsky et al., 2024), a substrate we do not run. Their objectives are anchored at an output, and circuits for one task can differ across its prompts (Makou et al., 2026).

**Circuits for internal features.** Internal targets are not new. Hierarchical attribution extracts the subgraph behind a chosen SAE feature (Ge et al., 2024), dictionary features have been traced patch-free in Othello-GPT (He et al., 2024), transcoder features are reverse-engineered from their input circuits (Dunefsky et al., 2024), latent attribution debugs a chosen latent (Dupre la Tour and Mossing, 2025), CircuitLens explains a feature's activation from upstream interactions (Golimblevskaia et al., 2025), and attribution graphs have been extended to concept directions read by linear probes (Palit et al., 2026). Upstream features have also been ranked by single ablations (Cunningham et al., 2024) or linked by sparse regression (Yin et al., 2026). These methods attribute, ablate one node at a time, or regress; circuit probing is closer in spirit, optimising a mask over model weights for a labelled intermediate variable (Lepori et al., 2023), and sparsely connected transcoders learn latent-to-latent masks while the dictionary is trained (jacob_drori, 2025).

**Masks and scalars.** Learned masks descend from subnetwork probing (Cao et al., 2021; Conmy et al., 2023) and edge pruning (Bhaskar et al., 2024; Li and Janson, 2024); over SAE latents they have localised attributes (Chaudhary and Geiger, 2024) and found behavioural circuits (Caples et al., 2025), the latter being our unweighted circuit masking with its objective moved to a latent. Such masks keep a node at its natural scale or ablate it; learned per-unit scales serve other jobs, in pruning, steering, and regression (Liu et al., 2017; Kwon et al., 2022; Stoehr et al., 2024; Yin et al., 2026), and gate-times-magnitude parametrisations separate membership from strength outside circuit discovery (Louizos et al., 2018; Lai et al., 2025).

**Evaluating circuits.** What removing a latent means can decide the result (Miller et al., 2024; Mueller, 2024); optimal ablation learns it per component (Li and Janson, 2024), the dual of our per-node coefficients, while our ablations stay fixed by data; denoising and noising answer different questions (Heimersheim and Nanda, 2024), and our sufficiency to induce is a denoising-style test. Prior work evaluates under several ablations but discovers under one; the nearest objectives sum two

directions of one counterfactual (Chen et al., 2025) or certify bounded patching perturbations (Hadad et al., 2026). Size-matched random circuits (Shi et al., 2024; Seth et al., 2026) go unfitted, like the circuits they test, while ours share our circuits' coefficient fitting; causal scrubbing (Chan et al., 2022) is stricter (Appendix K), and MIB standardises evaluation for output targets (Mueller et al., 2025).

What this paper adds is the conjunction: circuits of SAE latents *optimised* for a single target latent, with a gate and a separate *fitted* coefficient per node, trained under three ablations at once, and tested on held-out contexts against random circuits given the same fitting, for ~15k sampled latents per model [*pending: only TuringLLM so far; Pythia-70m and GPT-2 small, DAN-40/41*] (further related work in Appendix N).

# 3 Setting

The primary model is TuringLLM-1.0-254M, a 12-layer decoder-only transformer trained from scratch on a documented synthetic corpus of ~2 billion tokens generated by Phi-3 Mini (Abdin et al., 2024); a model whose training data is known is a more tractable subject than one trained on an opaque crawl (Appendix A). A bank of 36 Top-K sparse autoencoders, one per site (attention output, MLP output, and residual stream at each layer), maps each activation onto 40,960 latents with $K = 128$ active per position (Appendix A.4). The same protocol is applied to Pythia-70m and GPT-2 small with public Top-K dictionaries (Section 5.1). As a running example we follow a layer-9 residual-stream latent of TuringLLM that fires where a passage places its subject in classical Greek antiquity: after "In the 5th century BC, the philosopher taught in the city of" it reaches 76% of its typical activation, and after "In the 19th century, . . . " it stays silent (Section 5.3).

# 4 Method

For a latent $f$ and a sequence–position pair $(x, t)$, write $a_f(x, t) \geq 0$ for its activation (after Top-K) and $\hat{a}_f(x, t)$ for its pre-activation (before it); pseudocode is in Appendix P.

## 4.1 Problem

**Targets and contexts.** Fix a target latent $f$ and let $U_f$ be the latents at every SAE site that causally precedes $f$'s site. Each target has *activating contexts* from two stores, up to 64 of its strongest-activating sequences and up to 64 drawn from the middle of its activation distribution, each with an anchor at the target's peak (Appendix Q). With full stores it trains on 32 strongest and 16 mid-band contexts and is scored on 16 held-out strongest contexts (primary) and 16 held-out mid-band contexts (reported); the split is stratified by activation and always trains on the two strongest (Appendix B). Held-out strongest contexts exclude every training-side context and its exact, anchor-prefix and near copies (a shared run of $\geq 32$ tokens or $\geq 50\%$ shared 8-grams), filled to 16 from the next strongest unseen contexts (floor 8); held-out contrast contexts follow the same rule (Appendix B.1). It also has 64 *contrast contexts*: corpus sequences ranked by their highest cosine similarity to any of its strongest-activating contexts in pooled final-layer representation space, excluding its activating contexts, and kept only if the target stays out of its site's Top-K on them, so that they are close to the target's contexts but verified silent (Appendix Q). For the running example they are history passages about other periods, such as the Enlightenment, Renaissance music, or the pre-Columbian era. Targets are drawn uniformly at random, the same number per site (before targets without usable contexts are dropped), with no prioritisation heuristic (Appendix B).

**Circuit-only execution.** A *weighted circuit* is a node set $C \subseteq U_f$ with a scaling coefficient $\alpha_{s,i} > 0$ for each node. Under an ablation method $\pi$, the forward pass is modified at every upstream site $s$: the site's Top-K code $c_s$, computed from the (already modified) stream, is replaced by

$$\hat{c}_{s,i} = \begin{cases} \alpha_{s,i}\, c_{s,i} & (s,i) \in C \\ v^{\pi}_{s,i} & \text{otherwise,} \end{cases}$$

and the stream continues as $x + W^s_{\text{dec}}(\hat{c}_s - c_s)$, so SAE errors pass through unchanged and every downstream site, including the target's, reads the modified stream. Nodes are *recomputed*, not clamped: a node's value at each position is whatever the modified stream gives it. The ablation value $v^\pi$ is zero (Z), the latent's mean over activating contexts (A), or its mean over contrast contexts (C), optionally sparsity-preserving, with the per-position Top-K re-imposed. An unweighted circuit fixes $\alpha = 1$. Writing $a_C$ for the target's mean activation at the anchors under this intervention, $a_\varnothing$ for the empty circuit, and $a_{\text{pos}}$ for the unmodified model, *faithfulness* is

$$\phi^{\text{free}}_\pi(C) = \frac{a_C - a_\varnothing}{a_{\text{pos}} - a_\varnothing}.$$

**The problem.** Faithfulness depends on the ablation method (Miller et al., 2024; Mueller, 2024), and a circuit trained under one method can fail under another (Appendix T.3), so one weighted circuit is asked to serve all three:

$$n_\varepsilon(f) \;=\; \min\big\{\, |C| \;:\; \exists\, \alpha \;\text{ with }\; \phi^{\text{free}}_\pi(C, \alpha) \geq 1 - \varepsilon \;\text{ for every } \pi \in \{\text{Z}, \text{C}, \text{A}\} \,\big\}. \tag{P}$$

WCM solves a differentiable relaxation of (P); with $\alpha = 1$ the smallest faithful node sets are several times larger (Section 5.2). Problem (P) asks for *a* minimiser, and its near-optimal solutions form a large equivalence class (Appendix O), so only class-invariant quantities are treated as properties of the target.

## 4.2 Weighted Circuit Masking

Rather than thresholding attribution scores (the rank-and-cut baselines it is compared against, Appendix C), WCM learns the circuit directly, adopting the optimisation framing of learned-mask circuit discovery (Bhaskar et al., 2024; Caples et al., 2025) with a continuous-sparsification gate (Savarese et al., 2020), and moving its objective from an output to a single latent. For every upstream site $s$ it attaches gate parameters $\theta_s \in \mathbb{R}^{d_{\text{sae}}}$ with gates $m_s = \sigma(\theta_s)$, and coefficient parameters $\psi_s$ with $\alpha_s = \text{softplus}(\psi_s)$. A latent is kept at $\alpha$ times its live value when its gate is open and sent to its ablation value when closed:

$$x \;\leftarrow\; x + W^s_{\text{dec}}\big[\, m_s \odot \alpha_s \odot c^s + (1 - m_s) \odot c^s_{\text{abl}} - c^s \,\big],$$

where $c^s$ is the site's dense Top-K code and $c^s_{\text{abl}}$ its ablation value, a dense mean in training (the sparsity-preserving variant is used only in evaluation); editing only the decoded difference leaves the SAE error untouched, as in evaluation.

**Objective.** The circuit is trained to *reproduce* the target's pre-activation at the anchors of the training contexts, by mean squared error against its value on the unmodified stream. The pre-activation is forced by Top-K: the activation is censored at zero, so a loss on it carries no gradient exactly where the mask must learn. Reproducing the pre-activation does not, however, keep the target in its site's Top-K: the loss constrains the site's stream along one direction of $d_{\text{model}}$, and a circuit can restore that projection while other latents at the site rise past the target. A *rank-keep* term therefore asks the target to hold its natural rank: with $r$ the target's rank among its site's other latents at an anchor in the unmodified model, capped at $K$, and $\tau^{(r)}_\pi$ the $r$-th largest of their pre-activations under the circuit,

$$\mathcal{R}_\pi = \text{mean}\Big[\text{relu}\big(\tau^{(r)}_\pi - \hat{a}_f\big)^2\Big],$$

which is zero whenever the target is at or above its natural rank, so it never pulls a circuit towards a state the unmodified model does not have.

**Training under three ablation methods.** A mask trained under one ablation method learns whatever that method rewards: trained only under mean ablation over contrast contexts, its zero-ablation faithfulness is 0.0 at depth (it learns the difference from the contrast-context mean, never a set that suffices alone), and trained only under zero ablation, it is never asked what separates firing from a near-identical context that does not fire. WCM therefore scores one mask under all three ablation methods at once, with gradients accumulating into the same parameters:

$$\mathcal{L} = \frac{\sum_\pi \gamma_\pi \left(\mathcal{L}_\pi + \gamma_{\mathrm{R}}\, \mathcal{R}_\pi\right)}{\nu} \;+\; \lambda \sum_s \sum_i \sigma(\theta_{s,i}), \qquad \nu = \mathrm{mean}\left(\mathrm{target}^2\right),$$

summed over $\pi \in \{\mathrm{Z}, \mathrm{C}, \mathrm{A}\}$, with $\gamma_{\mathrm{Z}} = 1$, $\gamma_{\mathrm{C}} = \gamma_{\mathrm{A}} = 0.25$ at every layer, and $\gamma_{\mathrm{R}} = 3 \times 10^{-3}$; the ablation means are computed on training-split contexts only (Appendix B). All terms measure the same quantity on the same target, so one shared, bounded normaliser suffices; the sparsity penalty is flat per latent, and the calibration of the three-method design is in Appendix T.

**Scaling coefficients.** Coefficients start at $\alpha = 1$, so training starts exactly at unweighted behaviour; *unweighted circuit masking* keeps them there. One guard is required: with coefficients free, the optimiser can hold a gate just below the keep threshold while inflating its coefficient, emptying the node set while keeping function intact, so latents with closed gates are charged $\lambda\,(1 - \sigma(\theta))\,|\alpha - 1|$. The zero-ablation term anchors the coefficients: trained without it, they calibrate against the mean ablation's background and overshoot by one to three orders of magnitude once that background is removed (pre-v1; Appendix T.3).

**Binarisation.** Evaluation scores the hard node set $\{i : m_i > \frac{1}{2}\}$, so the gates are annealed during training, $m = \sigma(\theta/T)$ with $T$ decaying geometrically from 1 to 0.05 (continuous sparsification; Savarese et al., 2020), and the training forward converges to the binary gates that evaluation scores; the gates are optimised directly rather than sampled (Louizos et al., 2018; Li and Janson, 2024). Training runs 400 AdamW steps; all settings are in Appendix B, and the binarisation study in Appendix T.

## 4.3 Evaluation

Every reported score is read on held-out contexts and reads the target's activation (after Top-K) in every term of its ratio; the full evaluation family and its settings are in Tables 2 and 3.

**Intervention scores.** Let $A^+$ and $A^-$ be the target's mean activation on activating and contrast contexts. **Sufficiency to induce**, $\phi_{\mathrm{ind}} = (A^-_{\mathrm{int}} - A^-)/(A^+ - A^-)$, sets the circuit's nodes in contrast contexts to their mean values at the anchors of the activating contexts, times $\alpha$ for a weighted circuit ($\phi^{\alpha}_{\mathrm{ind}}$, the form reported throughout): near 1, the circuit switches the target on. **Necessity** is one minus the completeness score of Marks et al. (2025), with SAE errors preserved and the nodes zero-ablated: with $a_{M\setminus C}$ the target's activation on activating contexts when the circuit's nodes are set to zero and every other latent is recomputed,

$$\phi_{\mathrm{sup}}(C) = \frac{a_{\mathrm{pos}} - a_{M\setminus C}}{a_{\mathrm{pos}} - a_{\varnothing}}.$$

Both scores intervene on the circuit as one set. All scores are unclipped ratios: success means landing near 1, and a score far above 1 is overshoot, not quality.

**Ablation methods.** The choice of ablation is load-bearing (Li and Janson, 2024; Miller et al., 2024). Mean ablation over activating contexts leaks target signal into the ablation value increasingly with depth ($a_{\varnothing}$ up to 30% of $a_{\mathrm{pos}}$; Appendix R.2), while mean ablation over contrast contexts does not; dense means put the model on streams a Top-K SAE can never emit, so evaluation re-imposes the per-position Top-K after applying them (sparsity-preserving; Appendix R.1).

**Random-circuit baselines.** Fitted coefficients could make any set look faithful, so every random circuit, matched to the weighted circuit's size site by site, receives the *same coefficient fitting* on its own nodes before scoring. Its members are latents active on the target's strongest training contexts or, in two harder nulls, latents active near the anchor and matched on frequency or contribution (Appendix H).

**Specificity and near-threshold targets.** Every score reads the target alone, so a circuit that restores the target by lifting its whole neighbourhood at the site passes them all; we report specificity (the target's rank, the latents lifted above the Top-K cut, and the faithfulness of co-firing *sibling* latents) as a measured property, not a criterion (Appendix J). A *near-threshold* target, whose median rank at the held-out anchors of the unmodified model is at least $K/2$, has an ill-conditioned activation read, since small shifts near the cut push it out of the Top-K; such targets stay in the headline pass rate and are also reported as a stratum.

**Pass criterion.** A circuit passes if, on the held-out strongest contexts and the activation read, its faithfulness under all three ablation methods (Z, and sparsity-preserving A and C) lies in $[0.8, 1.5]$ and its necessity is at least 0.9. The band is asymmetric by design: below 0.8 the circuit fails to restore its target, while an overshoot up to 1.5 still restores it, and whether that overshoot is specific to the target is measured for every circuit. Sufficiency to induce, specificity, and the random-circuit baseline are reported rather than required; targets with a vacuous zero-ablation denominator are excluded and counted, and an undefined A or C ratio counts as a fail (Appendix B).

# 5 Experiments and Results

**Setup.** All results use protocol v1 (Appendix B), on TuringLLM and on the two public models of Section 5.1. The *production run* fits WCM to 15,046 sampled targets at one sparsity penalty; three samples of it are re-scored with every coefficient set to 1 (1,746 targets), against size-matched random circuits (747), and after a refit (175; Appendix O). A *sweep* fits weighted and unweighted circuit masking to 190 targets across sparsity penalties (Section 5.2), also used by the attribution comparison and the harder nulls, and 280 targets are refitted under four contrast-context sources (Appendix I). The TuringLLM numbers below were scored on the original held-out sets, which repeated training text (Appendix B.1) [*pending: re-scores on the rebuilt sets, DAN-157*].

## 5.1 Weighted Circuits at Scale

Figure 2 breaks the production run down by target layer and site: 15,004 fitted circuits (42 targets had too few activating or silence-checked contrast contexts, and 10 circuits with a vacuous denominator are excluded). Under the pass criterion (Section 4.3), 46% of circuits pass at the production sparsity penalty [*pending: rebuilt held-out headline and worst case with intervals, DAN-157*]. They hold a median of 494 upstream latents (386 for passing circuits), and 91% lie within $10^2$–$10^3$ (Figure 9). The pass rate depends on the target's site and depth: 65% of residual-stream, 52% of MLP, and 14% of attention-output targets pass, and the rate falls from 85% at layer 1 to 15% at layer 11. Depth breaks the mean-ablation background, not sufficiency: zero ablation and necessity hold for three circuits in four or more at every depth, while the share that fail only under the two mean ablations grows from 10% at layers 0–4 to 53% at 8–11. Attention outputs are the hard case: 40% of their circuits fail even zero ablation, mostly at targets that barely enter their site's Top-K. The full anatomy, transfer to weaker contexts, and sufficiency to induce are in Appendix G. Most circuits are specific to their target: under Z, A, and C the target is restored to a median 0.94, 0.99, and 0.92 of its pre-activation and its co-firing siblings to 0.26, 0.46, and 0.41, while a minority (30%) are flagged as amplifiers, which is why specificity is reported for every circuit (Appendix J).

**Random circuits.** Random circuits of the same size as each target's circuit, drawn from latents active on its strongest training contexts and given the same coefficient fitting, failed on all 747 targets

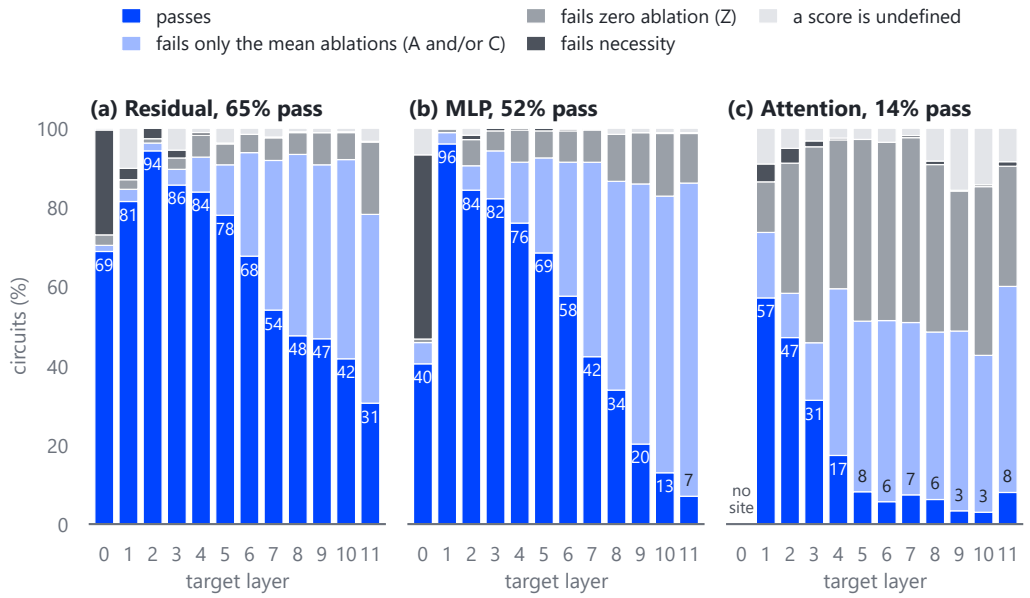



Figure 2: How the weighted circuits of 14,994 TuringLLM latents (the run's non-vacuous circuits) fare at the production sparsity penalty, by target layer and site kind. Each circuit is counted once: it passes (faithfulness under Z, A, and C in $[0.8, 1.5]$ and necessity $\geq 0.9$, on held-out contexts); fails only the mean ablations (A, C, or both) while zero ablation and necessity hold; fails zero ablation while necessity holds; fails necessity; or has an undefined score (3%, counted as a fail). Numbers on the bars are pass rates (%), and panel titles give each site kind's overall pass rate; layer-0 attention outputs have no upstream site.

of a stratified sample, against 43% for the fitted circuits on the same targets. Their median faithfulness is 0.00 under every ablation method at every depth, only one has even a single score in the pass band, and none induces the target to half its level in contrast contexts. This null is weak, however: its coefficients barely move (median exactly 0.798, the value weight decay alone reaches), and it fails on its training contexts too, so it shows that unrelated latents cannot be fitted, not that the selection matters; the harder nulls test that [*pending: harder-null pass rates on all three models, fitted and at* $\alpha = 1$*, DAN-181*].

**An operating point, not a ceiling.** The 46% is the pass rate at one sparsity penalty and one training budget. Weaker penalties give larger circuits that pass more often (31% of the 190 sweep targets at $\lambda = 4 \times 10^{-3}$, 58% at $2.5 \times 10^{-4}$, with the median circuit growing from 218 to 1,103 latents; Section 5.2), and at the 400-step budget most circuits are still shrinking (Figure 11), so reported sizes are sizes at a fixed budget.

**Beyond TuringLLM.** The same protocol is applied to two public models with public *Top-K* dictionaries: EleutherAI's $k$=16, 32,768-latent dictionaries for Pythia-70m (Biderman et al., 2023) and OpenAI's $k$=32, 32,768-latent dictionaries (Gao et al., 2024) for GPT-2 small (Radford et al., 2019), at every layer (sites in Table 3). Only the model and dictionary adapters differ; discovery, the ablation methods, the contrast contexts, and every score run through the same code as on TuringLLM. Each model gets 512 random targets, a sparsity penalty set by a size-only rule stated before a pilot, and TF32 fits (frame, corpora and rule in Table 3). [*pending: Pythia-70m and GPT-2 small: adopted* $\lambda$*, pass rate with interval and median size by depth band and site kind, the* $\alpha = 1$ *re-score, unweighted masking on Pythia-70m, DAN-40/41*]

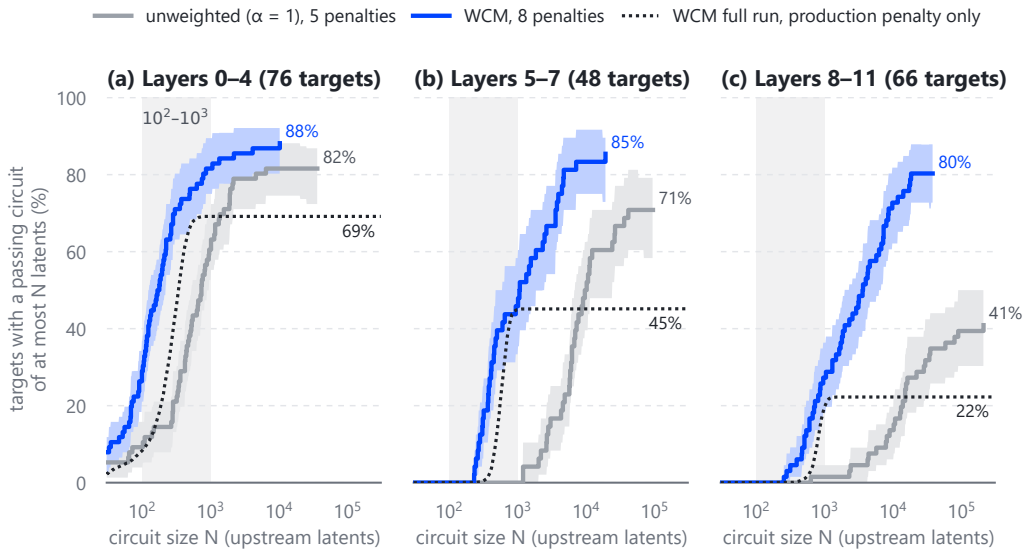



Figure 3: Fitted coefficients explain more targets at every circuit budget. For each target depth band, the share of 190 sweep targets with a circuit of at most $N$ upstream latents that passes the criterion of Section 4.3 (faithfulness under Z, A, and C in $[0.8, 1.5]$ and necessity $\geq 0.9$, on held-out contexts), taking each target's smallest passing circuit over the sparsity penalties it was fitted at: weighted circuit masking (blue) at eight penalties, $\lambda$ from $4 \times 10^{-3}$ to $2.5 \times 10^{-5}$, and unweighted circuit masking ($\alpha = 1$, grey) at five, $10^{-3}$ to $10^{-5}$. Targets that never pass count as unexplained; shading is a 90% bootstrap interval over targets, and each curve ends at the largest circuit fitted. Dotted: the full run (~15,000 targets) at the production penalty alone, which levels off at that penalty's pass rate.

## 5.2 Why the Coefficients Matter

Three versions of each circuit separate what the nodes contribute from what the coefficients contribute: the weighted circuit as fitted, the same nodes at natural scale ($\alpha = 1$), and a random circuit of the same size given the same coefficient fitting (Section 4.3). On 1,746 circuits sampled from the production run, the weighted circuits pass for 44.7% of targets and the same nodes at natural scale for 1.5%: median faithfulness falls from 0.94, 0.95, and 0.89 to 0.00, 0.09, and 0.06 under Z, A, and C, and at natural scale no depth band passes more than 4% of its targets (Figure 14). Random circuits given the same coefficient fitting pass for none (Section 5.1), so the fitted coefficients are necessary but not sufficient: they make a selected node set faithful, not an arbitrary one. How the coefficients act, by compensating for the ablated environment rather than restoring members to their natural values, is in Appendix H. Unweighted circuit masking isolates what the coefficients buy: the same objective and optimiser with $\alpha$ held at 1. Figure 3 follows 190 targets across a sweep of sparsity penalties and asks, for a budget of $N$ latents, what share of targets have a passing circuit that small. Weighted circuits explain more targets at every budget, and keep explaining more as the budget grows: within $10^3$ latents they pass for 82%, 48%, and 27% of targets at layers 0–4, 5–7, and 8–11, against 63%, 0%, and 2% for unweighted circuits, and over the whole sweep for 88%, 85%, and 80%, against 82%, 71%, and 41% even with unweighted circuits of up to $\sim 2 \times 10^5$ latents. Where both methods pass, the smallest unweighted circuit is larger by a median per-target factor of 3.7× at layers 0–4, 9.6× at 5–7, and 10.5× at 8–11, where it holds $\sim 10^4$ latents (median 13,075 against 906). The production penalty is one point on the weighted curves: the full run, fitted at that penalty alone, levels off at 69%, 45%, and 22% (dotted), while weaker penalties carry the same targets much further. At that shared penalty unweighted circuits pass 54% against 64% at layers 0–4 but 2.1% and 1.5% against 38% and 18% at layers 5–7 and 8–11 (Appendix H).

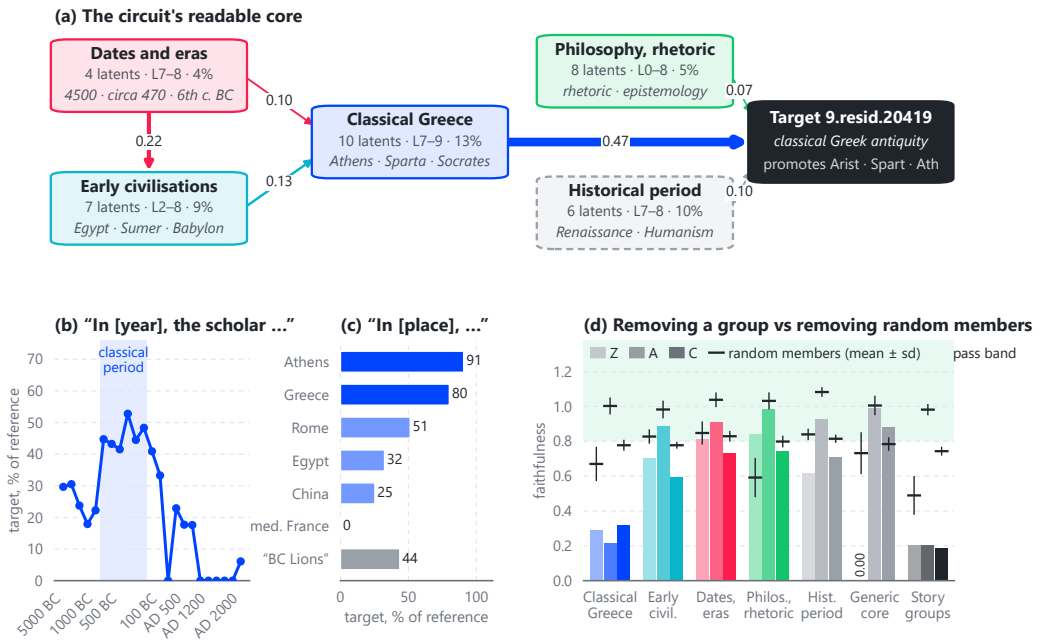



Figure 4: The weighted circuit of a latent for classical Greek antiquity (layer 9, residual stream; Section 5.3). **(a)** The readable core: role groups among the circuit's 60 largest contributors, named from their latents' own top contexts, with each group's size, layers, and share of the circuit's contribution at the target's anchor. Arrows give the drop in the destination when the source group is removed, under activating-mean ablation; the two left-hand groups also lower the target directly, by 0.11 and 0.12. The dashed grey group fires about as much on contrast contexts. Not drawn: a generic core of 7 latents and 682 further latents, together 55% of the contribution. **(b,c)** The target in the unmodified model on template sentences, as a percentage of its mean activation on its held-out strongest contexts. **(d)** Faithfulness with each group removed (bars) against removing the same number of random members of similar contribution rank (mean ± sd over 8 draws).

For the running example, the weighted circuit passes with 728 latents, while unweighted circuit masking on the same target fails at every penalty we tried, up to 24,837 latents, where its activating-mean faithfulness is still 0.71.

**Attribution at matched size.** On the 190 sweep targets, Appendix D compares WCM with attribution patching (AtP), integrated gradients (IG), hierarchical attribution, and ranked members given fitted coefficients, at every size each produces; against exact patching, AtP ranks latents better than IG (median rank correlation 0.86 against 0.71) [*pending: verdict, DAN-159 and DAN-77*].

## 5.3 A Circuit for Classical Antiquity

The running example shows what a passing circuit contains. Its target, a layer-9 residual-stream latent, fires where a passage places its subject in classical Greek antiquity, and its logit effect promotes "Arist", "Spart", "Ath", and "Greek". It tracks era and place rather than a word: completing "In [year], the scholar lived in the city of", it reaches 41–53% of its typical activation for 800–200 BC, 18–31% for older dates, and at most 6% from AD 1000 on, and for places it gives ancient Athens 91%, Rome 51%, Egypt 32%, and medieval France 0% (Figure 4b,c).

Its weighted circuit holds 728 latents and passes in the production run (Z, A, C 1.00, 1.14, 0.91; necessity 1.00); the knock-outs below are re-scored on a second GPU type, where the full circuit reads

0.98, 1.11, and 0.90. Grouping its 60 largest contributors by their own top contexts gives a readable core (Figure 4a). Ten latents at layers 7–9 respond to classical Greece itself (Athens, Sparta, Socrates, Thales) and carry 13% of the circuit's contribution; removing them takes the target to 0.29, 0.22, and 0.32 under Z, A, and C, against 0.67, 1.00, and 0.78 when as many random members of similar contribution rank are removed (Figure 4d). Upstream, latents for dates and eras feed latents for early civilisations (Egypt, Sumer, Babylon), and both feed the classical group: removing the dates group lowers the early-civilisation group by 22% and the classical group by 10%. Latents for philosophy and rhetoric, the subject matter of the target's training contexts, matter little. Removing all 21 latents of the three story groups, 3% of the circuit, takes the target to 0.20, 0.20, and 0.19 (random: 0.49, 0.98, 0.74).

The core is a reading, not the whole circuit: the story groups carry 27% of the contribution and 682 further latents most of the rest, and "BC" alone partly triggers the target ("The BC Lions" reaches 44%). The circuit is also flagged as an amplifier under activating-mean ablation, restoring co-firing latents almost as well as the target, although it lifts only 104 latents above the Top-K cut against a run median of 578 under the same ablation (Appendix J). Five further cases, including a latent for names that carry a physical law, are in Appendix L; these were found by searching, so Appendix M adds circuits for nine randomly drawn targets, passing and failing.

# 6 Discussion and Limitations

**Circuit size is a measurement at a stated scaling convention.** At natural scale a latent's circuit is large for deep targets (Section 5.2), and in (pre-v1) attribution node sets every latent is individually removable at the noise level while bulk removal craters the target (Appendix F). A latent's activation, reached through an additive residual stream, is a dense function of many small contributions, so any rule that thresholds individual contributions undercounts it however the threshold is tuned. We interpret most of that collective tail as redundancy that fewer, rescaled latents can carry, since fitted coefficients remove it (Figure 3); a circuit's size is therefore meaningful only alongside whether its coefficients were fitted.

**Ablation methods are part of the method.** Miller et al. (2024) showed that faithfulness is not robust to the choice of ablation; our measurements push the point into training. The home-turf effect (Appendix D), the contrast contexts behind the C ablation changing how far a circuit generalises (Appendix I), the miscalibration of coefficients trained without zero ablation (Section 4.2), and circuits that fail a resample test until trained against one (Appendix K) all say the same thing: the ablation methods used for discovery, training, and evaluation belong in the method statement. Equivalence classes of node sets, and composing circuits into chains, are discussed in Appendix O.

**Node sets are samples; function is stable.** Refitting a target with its training contexts in a different order, on a different GPU, gives a circuit that is about a fifth smaller, passes about as often, and shares a median 64% of the original's latents, and the shared core alone passes on 48 of the 56 targets where both fits pass (Appendix O); we therefore treat sizes and functional scores, not individual memberships, as properties of the target.

**Recommendations.** For work that discovers circuits for SAE latents, our results suggest five practices. Report faithfulness under more than one ablation method, and name the methods used for discovery, training, and evaluation, since rankings change between them (Appendix D). Compare circuit sizes only at matched faithfulness, and state whether coefficients were fitted. Report specificity alongside faithfulness, since a faithful circuit can be an amplifier. Report sizes, scores, and stable cores rather than a single membership, and expect bitwise reproducibility only on one GPU type (Appendix O). Keep near and anchor-prefix copies of training contexts out of held-out sets, not only exact copies (Appendix B.1).

### 6.1 Limitations

**The SAE basis is not ground truth.** Many latents resist clean characterisation, and circuits built from ambiguous latents inherit that ambiguity; our scores measure intervention effects, not interpretability.

**The objective is a surrogate.** WCM trains on the pre-activation, a continuous surrogate for a discrete Top-K event, and evaluation scores the activation itself; for near-threshold targets the gap is partly a scoring effect: in a sample their clean margin above the first excluded latent is 25% (against 81–91% for other targets), and under zero ablation their circuits reproduce the pre-activation (median 0.93) yet read 0 once the target slips out of the Top-K, though under the mean ablations they fall short on the pre-activation too (Appendix G).

**Attention-output targets pass least (14%).** An attention latent draws most of its signal from other token positions, while a circuit keeps one latent set at every position, so it must preserve the content of each source position; under mean ablation the failing circuits lose both where the dominant head attends and what it reads, and only larger circuits help (in a 22-target sample, about half pass at ~9k latents; Appendix G). Position-specific membership, or a separate latent set for the source positions, may lower that cost; we have not tested either.

**Circuits are fitted to one regime and one budget.** Circuits fitted mostly to strong contexts transfer only in part to weaker ones (12% pass on held-out mid-band contexts), and at the 400-step budget most are still shrinking, so pass rates and sizes describe a stated operating point (Appendix G, Figure 11).

**Faithfulness holds under the ablations a circuit was trained against.** When everything outside the circuit takes its value on another real context, as in causal scrubbing, 0.1% of circuits pass; training against such backgrounds raises this to 30% on 99 targets (Appendix K).

**Faithful at the anchor, not at every position.** Read at every position, deep circuits under zero ablation also fire on most contrast-context positions where the target is silent (74% at layers 8–11), so a circuit explains the target where it fires rather than simulating it everywhere (Appendix K).

**Faithfulness does not imply specificity.** A circuit can restore its target by lifting its whole neighbourhood; the rank-keep term limits the lifts that overtake the target but does not remove such circuits (30% are flagged), so specificity is measured rather than claimed.

**Contrast contexts.** Contrast contexts are verified silent only in that the target stays out of its Top-K, and "close" is one of several defensible constructions; circuits trained on close contexts pass less often when evaluated on distant ones than circuits trained on random contexts (Appendix I).

**Generality.** TuringLLM's synthetic corpus aids control but limits generality, which the two public models partly address; larger models remain open, and the method is scoped to Top-K dictionaries: on dense ReLU dictionaries it replicates, but a random-circuit baseline with fitted coefficients stops discriminating once a set spans the anchors (pre-v1, six targets; Appendix S).

## 7 Conclusion

We asked what makes an internal latent fire, and answered with weighted circuit masking. Two results carry the paper. Circuits for arbitrary latents can be found at scale and tested strictly: at a penalty that yields $10^2$–$10^3$ upstream latents, 46% of ~15,000 randomly sampled TuringLLM latents get a circuit that is faithful under three ablations and necessary on held-out text, while random circuits of the same size and fitting fail on all 747 targets tested. And fitted coefficients are what keep those circuits small: without them, deep circuits are about 10× larger, and half as many deep targets pass at any size we tried. Because the targets are sampled rather than chosen, the results describe a model's internal variables as a population, not a few tasks we thought to ask about; and because a circuit's size depends on whether its coefficients were fitted, a claim that some computation is dense or sparse needs its scaling convention stated. Circuits for single latents are a building block: composing them into chains, with propagation tested at every link, is the path from dependencies to a map of the model's internal computation.

# Reproducibility

The full system (pipeline stages, evidence stores, the discovery methods of Section 4, the target-intervention evaluation, and the analysis code that produces every figure and table in this paper) is available at `github.com/DanielJamesDavies/turing-explorer-circuit-discovery`. TuringLLM, its synthetic training corpus, and the trained SAE bank are described in Appendix A; every experiment runs from a released configuration under the protocol of Appendix B, and each analysis reads only the run artifacts written by the pipeline stages.

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# A Corpus and Model Details

## A.1 A Corpus Built by Design

The synthetic corpus was constructed in two stages. The first produced a relatively small but high-quality foundation: Phi-3 Mini (Abdin et al., 2024), a 3.8B-parameter language model available under a permissive licence, generated instructional and educational text spanning 18 broad subject areas (Physics, Mathematics, Computer Science, Chemistry, Biology, Philosophy, English, Creative Writing, Languages, Art, Music, Engineering, Economics, Business, Geography, History, Culinary Arts, and Communication), recursively generating topics and sub-topics and producing structured documents of headings and text.

The second stage addressed scale. A fine-tuned T5-based paraphrase model (Vorobev and Kuznetsov, 2023; Raffel et al., 2020) expanded the corpus by generating 11 paraphrased variants of each source document, motivated by knowledge distillation results (Ba and Caruana, 2014; Hinton et al., 2015) suggesting that varied outputs of a stronger teacher can be advantageous over raw data alone. The final corpus totals approximately 2 billion tokens.

Two tokenized datasets were derived using the Phi-3 Mini tokenizer. The language-modelling training set combines original and augmented texts shuffled into epochs, concluding with a second pass over only the high-quality originals to consolidate representations around the cleaner signal. The interpretability dataset consists of 64-token chunks shuffled into 3,030 shards, allowing the pipeline to process one shard at a time with a bounded memory footprint.

| Attribute | Value |
|---|---|
| Parameters | 254M |
| Layers | 12 |
| Residual stream dimension | 1,024 |
| Attention heads | 16 |
| MLP hidden dimension | 4,096 |
| Context length | 1,024 tokens |
| Vocabulary size | 50,304 tokens |
| MLP activation | SwiGLU |
| Normalisation | RMSNorm |
| Tokenizer | Phi-3 Mini |

Table 1: TuringLLM-1.0-254M architecture summary.

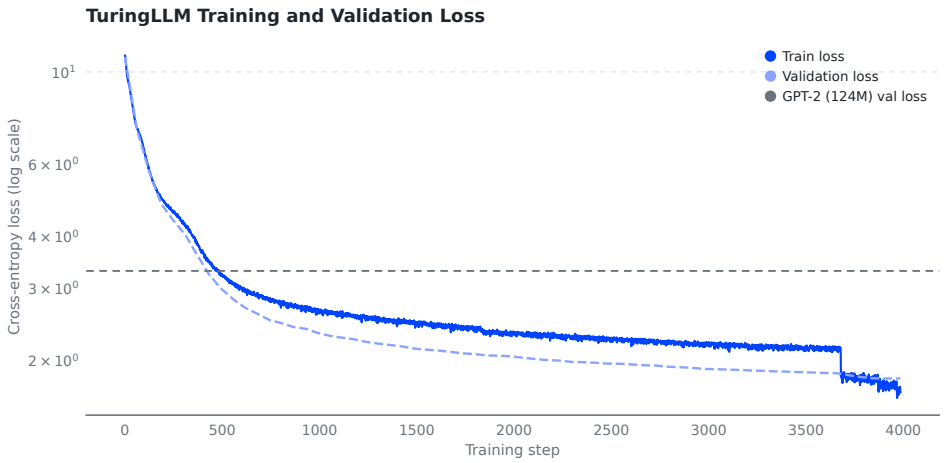



Figure 5: TuringLLM training and validation cross-entropy over the 3,985-step run (log scale). The dashed reference marks the validation loss reached by the same training infrastructure applied to GPT-2 (124M); the drop near step 3,700 corresponds to the final pass over only the high-quality Phi-3 Mini originals.

## A.2 Architecture and Training

TuringLLM follows the residual-stream design of the GPT family (Radford et al., 2019) with two deliberate departures: SwiGLU (Shazeer, 2020; Ramachandran et al., 2017) replaces the ReLU MLP, following the motivation of the Llama family (Touvron et al., 2023), and RMSNorm (Zhang and Sennrich, 2019) replaces layer normalisation. Table 1 lists its dimensions; SAEs attach at three sites per layer (attention output, MLP output, and the residual stream). Training reached a minimum cross-entropy of approximately 1.62 (train) and 1.80 (validation); the same infrastructure applied to GPT-2 (124M) reaches approximately 3.29 (Karpathy, 2024), reflecting the tighter distribution of the synthetic corpus. The loss curve (Figure 5) drops sharply in its final phase, corresponding to the second pass over Phi-3 Mini originals.

## A.3 Fitness for Interpretability

TuringLLM is not designed to win benchmarks; it is designed to be a controlled subject. Standard held-out benchmarks fit poorly for a base model trained on synthetic educational text, so fitness was evaluated qualitatively and through structured internal tests: Phi-3 Mini composed topic-specific opening sentences and scored TuringLLM's completions on English quality and topic adherence,

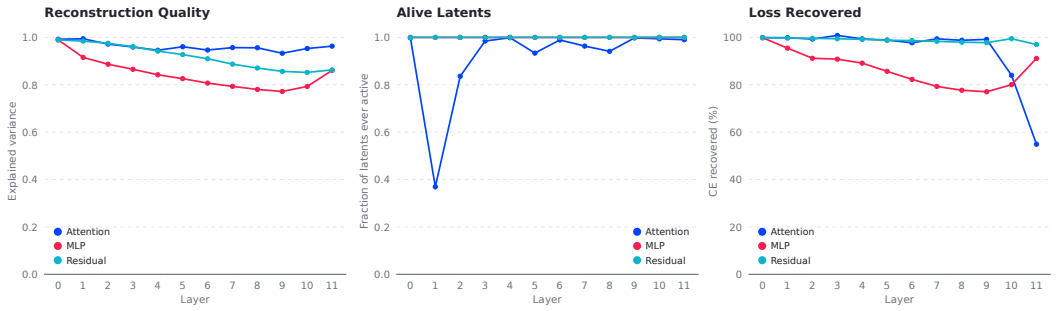



Figure 6: Quality of the 36-SAE bank by site. Left: explained variance of each SAE's reconstruction. Middle: fraction of latents that fired at least once over the full collection pass. Right: percentage of the cross-entropy gap recovered when splicing each SAE's reconstruction into the model, relative to zero-ablating the site.

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"Physics utilizes the fundamental concepts of quantum mechanics to explain how electrons populate a sublevel within an atom, which is marked by energy levels. These levels define the overall potential energy and enable electrons to occupy specific energy levels or sublevels [. . . ]"

"Artificial Intelligence (AI) has played a pivotal role in elevating personalization in human-robot interfaces, elevating the potential for truly personalized and responsive user experiences. The evolution towards AI extends its roots back to the 1960s with early rule-based [. . . ]"

"Artificial Intelligence (AI) and Machine Learning (ML) technology have become indispensable in modern engineering. These technologies enable engineers to analyze vast datasets with unprecedented efficiency, leading to significant improvements in design processes, maintenance, and optimization."

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Figure 7: Representative TuringLLM completions (excerpts). Continuations are topical and grammatical, supporting the use of the model as an interpretability subject; occasional repetitions and factual slips (e.g. conflated terminology) reflect its 254M-parameter scale and synthetic corpus.

yielding average grades of approximately B and A respectively across the 18 subject areas. This level of competence supports the working hypothesis that the model contains structured internal representations worth studying; a model merely interpolating noise would not produce consistently topical completions. Representative completions are shown in Figure 7.

## A.4 Sparse Autoencoder Quality

Figure 6 reports three standard quality metrics for the 36-SAE bank, per site. *Explained variance*, $1-\mathrm{Var}(x-\hat{x})/\mathrm{Var}(x)$, measures reconstruction fidelity. *Alive latents* is the fraction of each dictionary's 40,960 latents that fired at least once during the full collection pass over the interpretability corpus. *Loss recovered* splices each SAE's reconstruction into the model at its site and reports the recovered share of the cross-entropy gap between the original model and zero-ablating that site (Marks et al., 2025), measured over 12,288 tokens of the interpretability corpus. Because the SAEs are Top-K with $K = 128$, sparsity is fixed by construction and the usual $L_0$–fidelity trade-off curve is a single point per SAE.

Three observations matter for interpreting the paper's results. First, reconstruction fidelity is healthy across the bank: attention SAEs explain 0.93–0.99 of variance, residual SAEs decline smoothly with depth to 0.85, and MLP SAEs bottom out at 0.77 in layer 9. Second, the layer-1 attention SAE has a dead-latent crater: only 37% of its latents ever fire (84% at layer 2, at least 93% everywhere else). Third, loss recovered exposes a functional gap that explained variance misses: the layer-11 attention SAE recovers only 55% of the cross-entropy gap despite explaining 0.96 of variance (layer 10: 84%), indicating that the unexplained residual at the final attention site is disproportionately important to the logits.

# B Protocol

Every circuit in this paper is discovered and scored under the settings in Table 3, on every model; only the model and its dictionary change. Table 2 names the evaluation family; every reported number carries its row, is read on the held-out activating contexts (or held-out contrast contexts), and reads the target's activation in every term of its ratio.

| Score | Contexts | Circuit latents | Other latents |
|---|---|---|---|
| $\phi_{\text{ind}}$ | contrast | set | untouched |
| $\phi_{\text{sup}}$ | activating | zero | recomputed |
| $\phi_{\text{Z}}^{\text{free}}$ | activating | recomputed (permitted) | zero |
| $\phi_{\text{A}}^{\text{free}}$ | activating | recomputed | activating means, dense |
| $\phi_{\text{A,sp}}^{\text{free}}$ | activating | recomputed | activating means, sp |
| $\phi_{\text{C,sp}}^{\text{free}}$ | activating | recomputed | contrast means, sp |
| $\phi_{\text{Z},\alpha}^{\text{free}}$ / $\phi_{\text{A},\alpha}^{\text{free}}$ | activating | recomputed, $\times\,\alpha$ | zero / activating means |
| $\phi_{\text{ind}}^{\alpha}$ | contrast | set to $\alpha\times$ anchor means | untouched |

Table 2: The evaluation family. Intervention scores: $\phi_{\text{ind}}$ sufficiency to induce, $\phi_{\text{sup}}$ necessity; "set" means set, at every position, to the node's mean value at the anchors of the activating contexts. Faithfulness $\phi_{\pi}^{\text{free}}$ measures whether the circuit reproduces the target alone with its nodes recomputed, and each variant names its ablation method $\pi$ (Z zero, A mean over activating contexts, C mean over contrast contexts; sp sparsity-preserving; Section 4.1). The $\alpha$ rows score weighted circuits with their scaling coefficients applied (Section 5.2). SAE errors are preserved in every row, and every ratio reads the target the same way in all of its terms.

| Setting | Value |
|---|---|
| *Models and dictionaries* | |
| TuringLLM-1.0-254M | Top-K, $k = 128$, 40,960 latents; attention output, MLP output, residual stream at 12 layers |
| Pythia-70m | EleutherAI Top-K, $k = 16$, 32,768 latents; the same three sites at 6 layers |
| GPT-2 small | OpenAI Top-K, $k = 32$, 32,768 latents, layer-normalised input; attention output, residual stream between attention and MLP, MLP output, residual stream at 12 layers |
| Targets per model | TuringLLM: 455 drawn uniformly at random at every site with an upstream SAE site (layer-0 attention has none), 15,925 in all, restricted to the 15,046 that an earlier run of the pipeline could fit; this drops 879 attention-output targets that were dead or had fewer than 8 activating contexts, so attention sites hold 134–450 targets. 42 more are skipped for fewer than 8 strongest or 4 contrast contexts, leaving 15,004 circuits. Pythia-70m and GPT-2 small: 512 each, drawn uniformly at random with seed 20261024, 32 at each of Pythia-70m's 16 sites with an upstream site and 10–11 at each of GPT-2 small's 47, from latents that fire in at least 200 distinct windows of the model's corpus and in at most half of them (hub cap; the same cap would remove 1.75% of the TuringLLM targets); a disjoint pilot of 30 targets sets $\lambda$ |
| Corpora | TuringLLM: its interpretability corpus, 24.8M windows of 64 tokens (Appendix A); Pythia-70m: 4M windows from one pinned file of the uncopyrighted Pile (Gao et al., 2020); GPT-2 small: 4M windows of OpenWebText (Gokaslan and Cohen, 2019), as tokenised for its dictionaries; exact duplicate windows removed from the public corpora before the stores are built |
| Precision | TuringLLM fits and scores in bf16, its validated precision; Pythia-70m and GPT-2 small in float32 with TF32 matrix multiplies; context stores built in bf16, which only ranks contexts |
| *Contexts* | |
| Activating contexts | up to 64 strongest and up to 64 mid-band stored contexts, with repeated corpus ids removed but not repeated text (Section 4.1); a store with fewer than 8 is treated as absent. Each pool is split 3:1 into training and held-out contexts: ranked by anchor activation in blocks of 4, block $b$ holds out position $(b + 2) \bmod 4$, so ranks 1–2 always train. With full pools the target trains on 32 strongest (a stratified two-thirds of the 48 strongest training contexts) and 16 mid-band, and holds out 16 strongest (primary) and 16 mid-band (reported). Targets with fewer than 8 mid-band contexts train on all their strongest training contexts (665 of 15,004: 48 for 187, 6–47 for 478); in all, 502 targets train on fewer than 48 contexts and, as split, 472 had fewer than 16 held-out strongest contexts |
| Held-out rule | every held-out set excludes each training-side context (strongest and mid-band training contexts, all 64 contrast contexts, and every corpus copy of them) and its exact, anchor-prefix (identical up to the anchor) and near copies (a shared run of $\geq 32$ tokens, or $\geq 50\%$ of its 8-grams shared), and any copy of an already accepted held-out context; held-out activating contexts must fire at the anchor. TuringLLM: rebuilt for the fixed circuits; strongest filled to 16 from the original held-out contexts, then the strongest admissible contexts of a 512-context corpus scan; mid-band only filtered; floor 8 (below it a target is scored and flagged). Public models: pools split by whole groups of copies and same-document rows, then the rule applied as a final check (Appendix B.1) |
| Contrast contexts | 64 per target: candidates ranked by highest cosine similarity to any of the target's strongest stored contexts (pooled final-layer representations), its activating contexts excluded; the 8,192 most similar are scanned in order and kept only if the target is outside its Top-K at every position, until 64 are kept (Appendix Q); 48 train, 16 held out, stratified by similarity; the held-out 16 then go through the held-out rule against the contrast training contexts and every activating context, must stay silent at every position, and are refilled from the selector's next candidates. Targets with fewer than 4 are skipped; 310 have fewer than 64 |
| Anchor | position of the target's peak activation in each context |
| *Weighted circuit masking, WCM[Z+C+A]* | |
| Ablation methods | Z zero ablation (weight 1); C mean over contrast contexts ($\gamma_C = 0.25$); A |

## B.1 Held-out Contexts

The production run first built its held-out sets by the 3:1 split of each stored pool, and an audit of every record found that they were not clean: 66.8% of held-out strongest contexts repeated a training-side context, 35.6% exactly, 50.2% up to the anchor, and 55.1% as a near-duplicate. A context identical up to its anchor is the same input for every score, since each score reads the target at the anchor of a causal model. The cause is the corpus construction: about half of all 64-token windows (52.9%) were written to the interpretability corpus twice, the two copies have identical activations and so sit at adjacent ranks of a pool, and the split, which holds out one rank in every four, places most such pairs on both sides. The mid-band held-out sets were nearly clean (3.2% of contexts identical up to the anchor, 0.4% near-duplicates).

We therefore rebuilt every held-out set for the fixed circuits, under the held-out rule of Table 3; no circuit was refitted, and the training contexts, ablation means and contrast training contexts are unchanged. Under the rule a median of only 5 of the 16 original held-out strongest contexts survive, so a new scan of the corpus's 18.2M distinct windows ranked each target's 512 strongest contexts by the stores' own statistic, and the held-out strongest set keeps the surviving originals, then adds the scan's strongest admissible contexts until it holds 16. An admissible context must fire at its anchor, both when selected and when scored in the evaluation's own batch. Held-out mid-band contexts go through the same check and are only removed; held-out contrast contexts are checked against the contrast training contexts and every activating context, must stay silent at every position, and are refilled from the close selector's next candidates. Targets with fewer than 8 admissible strongest contexts are scored and flagged rather than dropped: headline pass rates are over targets with at least 8, and a worst case counts the rest as fails. [*pending: rebuilt held-out sets: targets below the floor by depth band and site kind, the share of scan contexts and their strength relative to the training contexts, headline and worst-case pass rates, DAN-157*] A paired re-score of a fixed stratified sample of 1,000 targets on the original and the rebuilt sets, on one machine, measures what the repetition did to the pass rate: [*pending: paired original against rebuilt pass rate on the 1,000-target sample, by depth band and site kind, DAN-157*]. For Pythia-70m and GPT-2 small, whose circuits are fitted after the rule was fixed, each pool is split by whole groups of copies (exact, anchor-prefix, and near-duplicate), so no group spans training and held-out contexts, and the rule is applied once more as a final check.

# C Attribution Baselines and Position-Aware Node Sets

WCM is compared against attribution methods that score every upstream latent and keep the top of a ranking. All of them run on each target's training contexts and score a latent's effect on the target's *encoder pre-activation* at the anchor,

$$\hat{a}_f(x,t) = \left\langle W_e^{(f)},\, z(x,t) \right\rangle + b_e^{(f)},$$

with $W_e^{(f)}$ the target's encoder row, $b_e^{(f)}$ its effective bias, and $z(x,t)$ the pre-SAE dense activation: unlike the activation, which Top-K censors to zero, it is defined on every forward pass, and gradients reach every upstream latent through identity passthroughs at each SAE site (Marks et al., 2025), with the SAE errors at their clean values.

## C.1 Estimators

The two gradient estimators share one path, from every upstream site's mean over the target's activating training contexts (its A ablation value, $\bar{c}^s$) to its natural code $c^s$, all sites interpolated together and held at their interpolated codes, and both multiply a gradient by $(c^s - \bar{c}^s)$:

- **Attribution patching** (AtP) takes the gradient at the clean point, giving the first-order estimate of mean-ablating each latent (Nanda, 2023; Syed et al., 2023; Kramár et al., 2024).
- **Integrated gradients from the mean** (IG) averages the gradient over 10 points of the path. It follows the $\mathrm{IE}_{\mathrm{ig}}$ node estimator of Marks et al. (2025), the node-level counterpart of EAP-IG (Hanna et al., 2024), in form but not in detail: theirs interpolates one submodule at a time

with the rest of the model live, starts from a zero patch when no counterfactual exists, and scores the error terms as nodes.

The two differ only in where they linearise, with target, contexts, path, and selection held fixed. Two further baselines use them:

- **Hierarchical attribution** applies the gating algorithm of Ge et al. (2024) to our SAE sites. Without transcoders our graph is not linear, so its scores are first-order gradients rather than their exact linear attributions, aggregated over contexts rather than per prompt; its threshold is bisected towards the weighted circuit's size, and every bisection point is scored.
- **Ranked members with fitted coefficients** fill the weighted circuit's per-site counts from the top of a ranking at each site, then fit coefficients exactly as for random circuits (Section 4.3: only the coefficients move, with no sparsity penalty), and are scored at $\alpha = 1$ and fitted. The rankings are IG, the latent's mean activation at the target's anchor over the training contexts (no gradient), and the correlation of that activation with the target's across contexts (co-activation). Against WCM, these rows separate what the selection contributes from what the coefficients contribute.

IG makes the attribution baselines comparable in kind to the published estimator, but not identical to it, so the output-targeted comparison of Appendix E uses its own reimplementation of the published recipe; a version of that recipe with the target latent as its metric, which would isolate the choice of target from everything else, is [*pending: DAN-48*].

**Exact activation patching.** Attribution patching tracks exact activation patching more closely than integrated gradients do. On ten targets (one per depth band and site kind, and a tenth at layers 8–11), we measured the exact effect of each candidate latent: the change in the target's pre-activation at the anchor when that latent alone is set to its training-context mean at every position and everything else stays live, averaged over the training contexts, which is the quantity AtP linearises. The candidates, 3,200–3,900 per target out of 8k–760k active latents, are the 2,000 top-ranked by each of AtP and IG and 1,000 other active latents drawn at random, so the comparison covers the heads of the two rankings and a random sample of the rest. Under the aggregate ranking defined below, the median rank correlation with the exact effect is 0.86 for AtP and 0.71 for IG (0.81 and 0.62 on magnitudes), and of the exact top-$n$ latents at the weighted circuit's size, AtP's top-$n$ holds a median 79% and IG's 54%; ranked by their largest score at any context and position, both track the exact effect less closely (0.72 and 0.62). These measurements use training contexts only. Members ranked by exact effect, given fitted coefficients, give a ground-truth row for the comparison: [*pending: exact-ranked circuits scored on the rebuilt held-out sets, DAN-159*].

## C.2 Rankings and Node Sets

Attribution returns a score for every latent at every context and position, so the baselines must decide how the scores form one ranking and which node set a ranking defines. Three rankings collapse contexts and positions: a latent's largest |score| at any context and position; the mean over contexts of its summed |score| over positions; and the magnitude of the mean over contexts of its summed signed score, the aggregate ranking of prior work (Marks et al., 2025). A truncated baseline keeps the top $n$ latents of one ranking, with $n$ the target's weighted-circuit size.

At natural size the Top-K constraint shapes the node set: the code is $k$-sparse *per position* while a target reads its whole causal prefix, so a position-collapsed circuit must cover the union of every position's active set, thousands of distinct latents per site, or recomputed faithfulness collapses. The baselines' node set is therefore a set of latents, the same kind of object (P) optimises over, collected across positions. Latents outside the circuit are ablated everywhere; nodes are not forced to any value: they are recomputed from the modified stream at every position, so positions instantiate naturally (allow, don't force), exactly the circuit-only execution of Section 4.1. Discovery is position-aware to match: attribution is computed per position, a percentile cut on |attribution| (pooled over sites and positions, resolved per batch of four contexts) keeps individual (latent, position) entries, and the node set is the union over the positions of the target's causal prefix. With a mean-ablation path nearly every

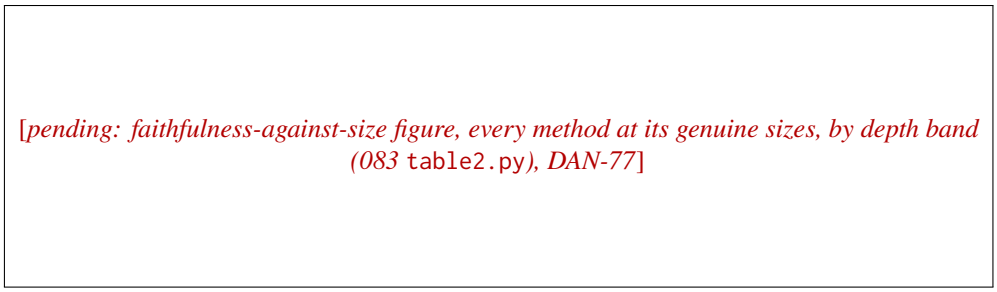



Figure 8: Pass rate (top) and median activating-mean faithfulness (bottom) against median circuit size, by target depth band, on the 190 sweep targets: weighted and unweighted circuit masking at each sparsity penalty, AtP and IG truncated to the weighted circuit's size at each WCM penalty (under the largest-score and aggregate rankings) and at their natural size, every bisection point of the hierarchical attribution, and the ranked-member baselines at $\alpha = 1$ and fitted.

entry is nonzero, so the median cut keeps about half of them, which is why these unions are large (the natural-size rows of Table 4). No template or cross-example alignment is required. Both attribution signs are kept, as a design principle rather than an implementation detail: the node set is *stream reconstruction*, not a selection of what drives the target, and a negative-attribution latent still carries part of the residual stream that nodes are recomputed from, so removing it corrupts recomputed execution.

A node set's size at a stated faithfulness is a measured quantity, and Appendix F argues it measures the structure of the computation rather than a defect of the method.

# D Method Comparison across Circuit Sizes

Every method is compared on the 190 sweep targets under protocol v1, scored on the rebuilt held-out sets (Appendix B.1) with the pass criterion of Section 4.3, by depth band. The comparison is a curve first (Figure 8): pass rate and median activating-mean faithfulness against median circuit size, with every method at the sizes it actually produces. WCM appears at its eight sparsity penalties and unweighted circuit masking at its five (Figure 3); AtP and IG are truncated to the weighted circuit's size at every WCM penalty and also shown at their natural size; every bisection point of the hierarchical attribution is shown; and the ranked-member baselines sit at the weighted circuit's per-site counts (Appendix C). Table 4 summarises the curve at one point per method: each method at its natural operating point, with WCM and unweighted circuit masking at the same sparsity penalty, $\lambda = 10^{-3}$, rather than at matched size, since no unweighted penalty on the grid gives circuits as small as WCM's (the nearest is $\lambda = 10^{-3}$ for 99% of targets, at a median 1.85× the WCM size); then AtP and IG truncated to the WCM size at $\lambda = 10^{-3}$ under each ranking; then the ranked members at $\alpha = 1$ and fitted. A pre-v1 version of this comparison, on a 22-target panel and with older attribution code, is retired; only its attribution-mass measurement, below, is kept.

Three questions structure the comparison. First, at natural size, does attribution reconstruct the target with its full per-position union, and do the methods' node sets coincide there, so that method identity matters only under pressure to be compact? [*pending: natural-size rows and node-set overlap, DAN-159*] Second, at the weighted circuit's size, does any attribution ranking pass, the aggregate ranking and the hierarchical attribution included, and how does unweighted circuit masking at the same penalty compare? On the pre-v1 panel every truncation failed under mean ablation and the weighted circuit passed on most targets. [*pending: truncated and unweighted rows by depth band, DAN-159*] Third, do fitted coefficients rescue members chosen by attribution, by activation, or by exact effect, which separates what the selection contributes from what the coefficients contribute? [*pending: ranked-member rows at $\alpha = 1$ and fitted, DAN-159 and DAN-43*] Throughout this programme, a method has tended to win the scores whose ablation method it was trained or ranked under, which we call the *home-turf effect*, and under the prior work's own first-order evaluation the pre-v1 ranking

| Method | $n$ (med) | $\phi_Z^{free}$ | $\phi_A^{free}$ | $\phi_C^{free}$ | $\phi_{sup}$ | pass (%) 0–4 | 5–7 | 8–11 |
|---|---|---|---|---|---|---|---|---|
| *Natural operating point* | | | | | | | | |
| weighted circuit masking, $\lambda = 10^{-3}$ | ? | ? | ? | ? | ? | ? | ? | ? |
| unweighted circuit masking, $\lambda = 10^{-3}$ | ? | ? | ? | ? | ? | ? | ? | ? |
| AtP, position union | ? | ? | ? | ? | ? | ? | ? | ? |
| IG, position union | ? | ? | ? | ? | ? | ? | ? | ? |
| hierarchical attribution, bisected | ? | ? | ? | ? | ? | ? | ? | ? |
| *Truncated to the weighted circuit's size* | | | | | | | | |
| AtP, largest score | ? | ? | ? | ? | ? | ? | ? | ? |
| AtP, mean \|score\| | ? | ? | ? | ? | ? | ? | ? | ? |
| AtP, aggregate | ? | ? | ? | ? | ? | ? | ? | ? |
| IG, largest score | ? | ? | ? | ? | ? | ? | ? | ? |
| IG, mean \|score\| | ? | ? | ? | ? | ? | ? | ? | ? |
| IG, aggregate | ? | ? | ? | ? | ? | ? | ? | ? |
| *Ranked members at the weighted circuit's per-site counts: $\alpha = 1$ / fitted* | | | | | | | | |
| IG-ranked, $\alpha = 1$ | ? | ? | ? | ? | ? | ? | ? | ? |
| IG-ranked, fitted | ? | ? | ? | ? | ? | ? | ? | ? |
| activation-ranked, $\alpha = 1$ | ? | ? | ? | ? | ? | ? | ? | ? |
| activation-ranked, fitted | ? | ? | ? | ? | ? | ? | ? | ? |
| co-activation-ranked, $\alpha = 1$ | ? | ? | ? | ? | ? | ? | ? | ? |
| co-activation-ranked, fitted | ? | ? | ? | ? | ? | ? | ? | ? |

Table 4: The method comparison at one point per method, on the 190 sweep targets (protocol v1, rebuilt held-out sets): medians of held-out faithfulness under zero (Z), activating-mean (A), and contrast-mean (C) ablation and of necessity, and the pass rate under the criterion of Section 4.3 by target depth band. Weighted-circuit and fitted rows apply their coefficients; the rankings and baselines are defined in Appendix C. [*pending: every cell, from experiments/083-v1-comparison* `table2.py`*, DAN-159; the exact-patching-ranked rows on ten targets as a separate block or a note, DAN-159*]

inverts (below); rankings are therefore a matrix of discovery method against ablation method, never a single column.

**Attribution-mass recovery (pre-v1).** Ge et al. (2024) score circuits by attribution recovery inside their linearised graph, with no intervention; our first-order analogue, measured on the retired pre-v1 panel, is *attribution-mass recovery*, the share of total activation-times-gradient mass a node set captures (a surrogate: the exact leaf-sum identity exists only in their substitute model). Under it that panel's intervention ranking inverts: plain top-$n$ attribution leads (median signed recovery 0.76), the replication follows (0.72), and the weighted circuit trails (0.52) while being the only arm that passed the intervention test there. The home-turf effect therefore reaches evaluation frameworks themselves: a circuit that reproduces the target on held-out intervention while holding a third of the absolute attribution mass (0.37 median) shows that first-order attribution mass is not what faithfulness is made of.

# E Output-Targeted and Latent-Targeted Circuits

**Output-targeted circuits do not explain latents, and vice versa.** A reimplementation of the sparse-feature-circuits recipe (Marks et al., 2025) (integrated gradients one submodule at a time from a zero patch, error nodes, an output metric) on the same model and SAE bank scores $\phi_{ind} = \phi_{sup} = 0.000$ on each of the three deep target latents it was run on (layers 8–10), even at its most permissive threshold, and our circuits score near zero on logit faithfulness (Appendix Q.1): "which latents support this output?" and "what makes this latent fire?" are answered by circuits that fail each other's objective. The output-targeted arm ran at its ceiling threshold, in-sample, and before protocol v1 and its evaluation fixes, so this is a scope result on three targets, not a ranking; [*pending: re-run under protocol v1, DAN-48*].

# F Individual Contributions Undercount Faithfulness

This appendix is pre-v1, and so are all of its numbers: it uses node sets from an earlier version of the attribution baselines (Appendix C) at natural scale ($\alpha = 1$), on one target each at layers 2, 8, 9, and 10, under the evaluation that preceded protocol v1. Leave-one-out is blind to how these node sets work: removing any single node changes the target by 0.000, the noise level. Yet cutting nodes in bulk by their own attribution ranking craters function: dropping the bottom half by score moves $\phi_Z^{\text{free}}$ from 0.975 to 0.895 at layer 2 but from 1.119 to 0.595 at layer 8; dropping the bottom 90% yields 0.635, 0.000, 0.222, and 0.005 at layers 2, 8, 9, and 10. The scores being cut are minuscule, $2\text{–}5 \times 10^{-4}$ in attribution units. Nodes individually at the noise level collectively carry the entire deep reconstruction. Individual-contribution thresholding, the standard selection rule across circuit discovery, therefore undercounts what faithfulness needs, which is why circuit size is reported here as a measurement at a stated faithfulness.

# G The Production Run in Detail

**Where the fall with depth comes from.** Zero ablation and necessity both hold for three circuits in four or more at every depth (79%, 80%, and 76% at layers 0–4, 5–7, and 8–11; Figure 2); what grows with depth is the share that fail only under the two mean ablations, from 10% of circuits at layers 0–4 to 53% at 8–11 (over all layer-11 circuits, median A and C faithfulness are 0.64 and 0.48), and these make up 59% of all failing circuits. Attention outputs are the exception: 40% of their circuits fail even zero ablation, and 63% of those belong to near-threshold targets, which barely enter their site's Top-K in the unmodified model and read 0 once they slip out of it (1,142 targets, 7.6% of the run, almost all attention outputs; 0.8% pass, against 50% for the remaining targets). Layer 0 departs from the trend (55%): its circuits are faithful (90%), but over a third (37%) fail necessity, because the token embedding reaches layer-0 targets directly and is never ablated.

**Transfer to weaker contexts.** On the held-out mid-band contexts, where targets fire more weakly, 12% pass (median largest deviation 0.77, against 0.31 on the strongest contexts), so circuits fitted mostly to strong contexts transfer only in part to weak ones: a circuit is a circuit for the regime it was fitted in.

**Why attention outputs fail.** Three local tests on samples of production attention targets locate the cost in source positions. About 85% of an attention target's clean contribution comes from source positions other than its own and the first token, for passing and failing targets alike. Failing targets read from more positions and more heads than passing ones at the same circuit size (layers 0–4: 4.5 against 2.4 effective positions, 7.1 against 4.5 effective heads). Under activating-mean ablation the dominant head of a failing circuit keeps 0.42 of its contribution when only its attention pattern is ablated and 0.60 when only its values are (passing circuits: 1.04 and 0.80), so the circuit loses both where the head attends and what it reads. Holding attention patterns at their clean values does not rescue them (5 of 120 failing circuits), and raising the A and C weights or the rank-keep weight barely helps (21% to 29% on 24 targets); at a weaker penalty about half pass, at ~9k latents (10 of 22 attention targets at $\lambda = 2.5 \times 10^{-5}$). Near-threshold targets are a separate case, largely a Top-K censoring effect: their clean margin above the first excluded latent is 25%, against 81–91% for other targets, and over all 1,142 of them the pre-activation read gives median faithfulness 0.93 under zero ablation but only 0.50 and 0.42 under the two mean ablations, so the read explains their zero-ablation failures but not all of their mean-ablation ones.

**Sufficiency to induce by site kind.** Sufficiency to induce, unlike reconstruction, is structured by the target's site kind (Figure 10). It asks a question no circuit is trained on: setting the circuit's latents to $\alpha$ times their mean values at the anchors of the activating contexts, does the target switch on in held-out contrast contexts where it is silent? Across the production run the median is 0.84 (0.98 among passing circuits, of which 88% switch the target on to at least half its natural level). Residual-stream targets

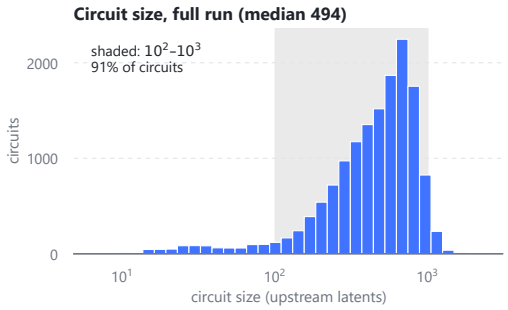



Figure 9: Circuit sizes in the production run at the production sparsity penalty: 91% of circuits hold $10^2$–$10^3$ upstream latents (shaded); median 494.

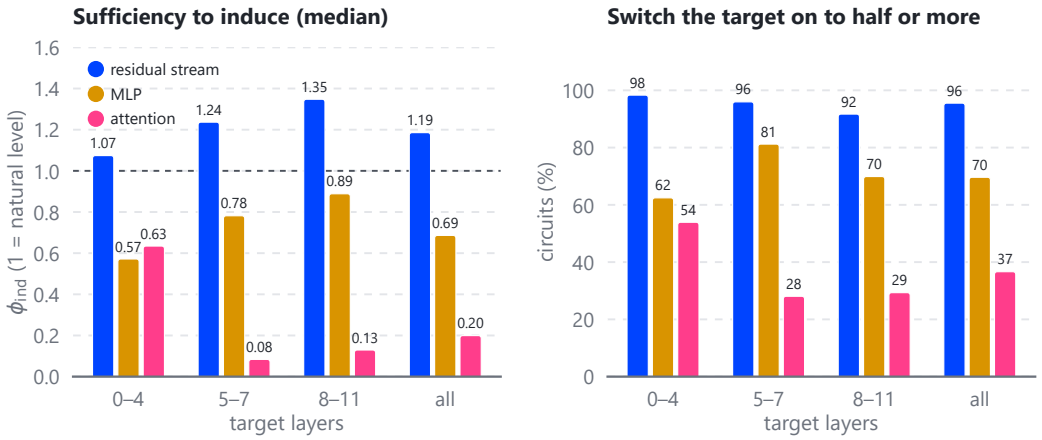



Figure 10: Sufficiency to induce across the production run (14,994 non-vacuous TuringLLM circuits), on held-out contrast contexts where the target is silent. Left: median $\phi^{\alpha}_{\text{ind}}$ by target depth band and site kind (1 is the target's natural level, dashed). Right: the share of circuits that switch the target on to at least half its natural level. Reported for every circuit; not part of the pass criterion.

are induced almost everywhere (96% at half or more; median 1.19, rising with depth to 1.35 at layers 8–11), MLP-output targets partly (70%; median 0.69), and attention-output targets rarely beyond layer 4 (28% and 29% at layers 5–7 and 8–11; median 0.20 overall). Which latents *can* be induced from upstream is therefore itself a per-kind property, and characterising it is open.

**Training dynamics.** Every production fit records its per-step curve (Figures 11 and 12). All gates start open, so no latent falls below the keep threshold for the first 57 steps; membership then collapses from $10^5$–$10^6$ candidates to a median of about 1,700 by step 100, and a circuit is within twice its final size by a median step of 137. It is not converged at the protocol's 400 steps: the median circuit loses 12% of its members over the last 100 steps (16–17% at layers 10 and 11), and 88% are still shrinking by more than 5%. The data loss stays nearly flat over that stretch while the penalty falls, so the late pruning costs little training fit, but at depth the circuits still shrinking pass less often: 30% against 67% for the settled half at layers 4–7, and 9% against 36% at layers 8–11, with no difference at layers 0–3. Reported sizes are therefore sizes at a fixed budget. The sparsity penalty dominates the loss throughout: its tail of nearly closed gates outweighs the members, and the data terms end at 0.3% of the loss. Before their 0.25 weights, the contrast-mean and activating-mean errors are about twice the zero-ablation error, the same ordering as held-out faithfulness.

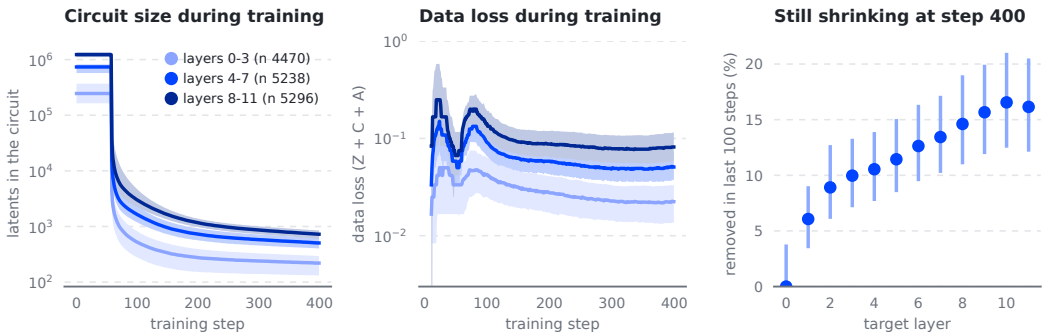



Figure 11: Training dynamics of the 15,004 production fits, by target depth band (layers 0–3, 4–7, and 8–11; median and interquartile range). Left: circuit size (latents above the keep threshold) per step. Middle: the data loss (the Z, C, and A terms, penalty excluded), averaged over each pass through the 48 training contexts (12 steps), since every step sees one batch of 4. Right: the share of members removed over the last 100 steps, by target layer (median and interquartile range).

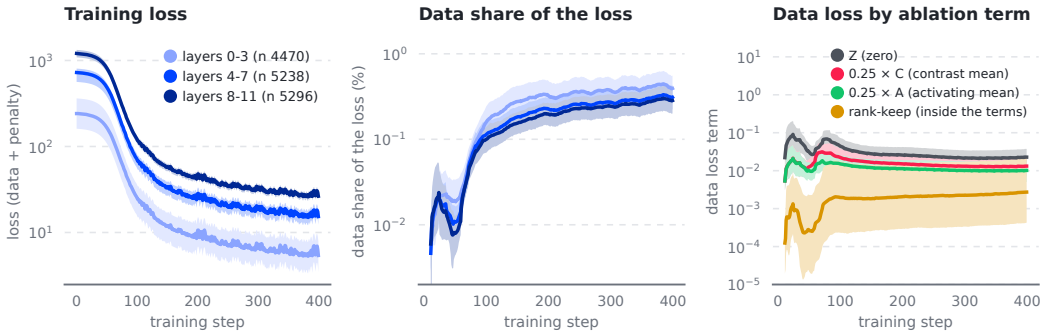



Figure 12: The training loss of the production fits (median and interquartile range; per-pass averages as in Figure 11). Left: the total loss, data terms plus sparsity penalty, by target depth band. Middle: the data terms' share of it; the rest is the penalty on open gates. Right: the data loss split into its ablation terms as weighted in the objective (Z, 0.25× C, 0.25× A), with the rank-keep contribution that each term contains.

# H Coefficients and the Weighted-versus-Unweighted Sweep

**Weighted against unweighted circuits across the sparsity penalty.** The comparison at a fixed penalty (Figure 13) uses the one value both grids share, $\lambda = 10^{-3}$; the rest comes from the full sweep, which fits WCM at $\lambda \in \{2.5, 5, 10, 20, 40\} \times 10^{-4}$ and unweighted circuit masking at $\lambda \in \{0.1, 0.3, 1, 3, 10\} \times 10^{-4}$ to 191 targets (five per layer and site kind, plus the pilot targets; 190 with usable contexts) and scores every circuit on the held-out strongest contexts. Figure 3 adds WCM at $\lambda \in \{0.25, 0.5, 1\} \times 10^{-4}$ for every sweep target, so that both methods reach circuits of $\sim 10^4$ latents and beyond. At the shared penalty unweighted circuits keep 1.85× as many latents and pass 54% against 64% at layers 0–4, but 2.1% against 38% at 5–7 and 1.5% against 18% at 8–11 (Figure 13): their scores are lower one by one, and the criterion asks for all of them at once.

Unweighted masking needs a weaker penalty to reach comparable faithfulness, so the two grids differ. Figure 17 separates the three ablation methods: the gap is smallest under zero ablation and largest under the two mean ablations, where unweighted circuits of $\sim 10^3$ latents score 0.18–0.37 at layers 5–11 (zero ablation 0.58–0.72). On the five-penalty grid WCM stops near $10^3$ latents, so the size ratios in the body come from the eight-penalty comparison of Figure 3, which adds three weaker

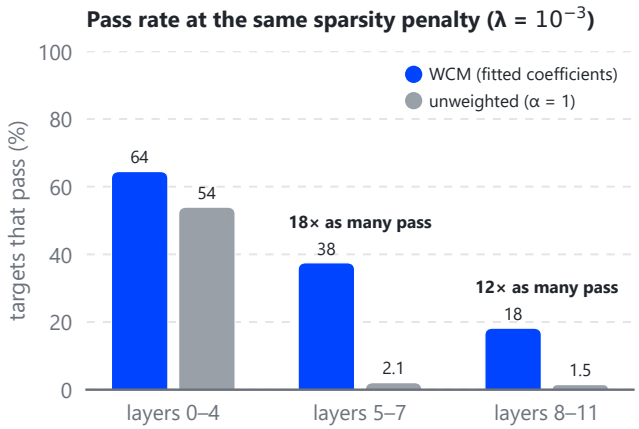



Figure 13: Weighted against unweighted circuit masking at the shared sparsity penalty, $\lambda = 10^{-3}$ (the production setting): the share of targets whose circuit passes the criterion of Section 4.3 (faithfulness under Z, A, and C in $[0.8, 1.5]$ and necessity $\geq 0.9$, on held-out contexts) for weighted circuit masking (blue) and unweighted circuit masking ($\alpha = 1$, grey), by target depth band; bold labels give the ratio of pass rates. 190 targets, protocol v1; the per-score medians are in Figure 16.

WCM penalties: per target, the smallest passing unweighted circuit is a median 3.7× (62 targets that pass under both methods), 9.6× (34), and 10.5× (26; median 906 against 13,075 latents) larger than the weighted one at layers 0–4, 5–7, and 8–11. Pooled over depth bands, WCM's pass rate rises monotonically as the penalty weakens: 31%, 35%, 42%, 47%, and 58% at $\lambda = 40, 20, 10, 5, 2.5\times10^{-4}$, where the median circuit holds 218, 327, 472, 747, and 1,103 latents; the production setting is one point on this curve, which has not levelled off.At the production penalty the per-score medians are all lower unweighted but not dramatically so at layers 8–11 (zero ablation 0.58 against 0.89, activating mean 0.27 against 0.69, contrast mean 0.18 against 0.51; Figure 16); the pass rates differ far more because the criterion needs every score at once.

**The coefficient triad.** Figure 14 sets the three versions of a circuit side by side: the circuits as fitted (the full run, 46% pass; 44.7% on the 1,746 targets of the next sample), the same nodes with every coefficient set to 1 (1.5% of 1,746; at most 4% in any depth band or site kind), and random nodes of the same size given the same coefficient fitting (0 of 747). At natural scale the nodes seldom overshoot the target: 4 of the 1,746 circuits exceed 1.5 under any ablation method, all at attention-output targets. More often they fall short of the empty circuit itself under activating-mean ablation (18%; 7% at layers 0–4, 32% at 8–11, and 29% of attention-output targets). There the activating-context means that the members replace already carry more of the target (a median 1.16 of a natural 3.06) than the members do at their natural scale (0.61), and the fitted coefficients make up the difference; these circuits pass less often with fitted coefficients (16%, against 51% for the other targets).

**Harder random circuits.** The random circuits of the triad are the weakest null that fits coefficients. Their members are latents active anywhere on the target's strongest training contexts, and most are silent at the positions that feed the anchor, so their coefficients receive no gradient: the median coefficient is exactly 0.798, the value weight decay alone reaches from 1 in 400 steps, and on the circuits' own training contexts the median zero-ablation faithfulness is 0.00 (0.4% reach 0.8). The null therefore shows that unrelated latents cannot be fitted, not that WCM's selection matters. Two harder nulls keep everything else and change only where the members come from. The *anchor-active* null draws, at each site, the production circuit's member count from latents active at the anchor or up to 8 positions before it on at least one training context, matching each member's corpus-frequency decile; production members may be drawn. The *contribution-matched* null also matches each member's contribution to the target, the mean over training contexts of its activation times the gradient of the

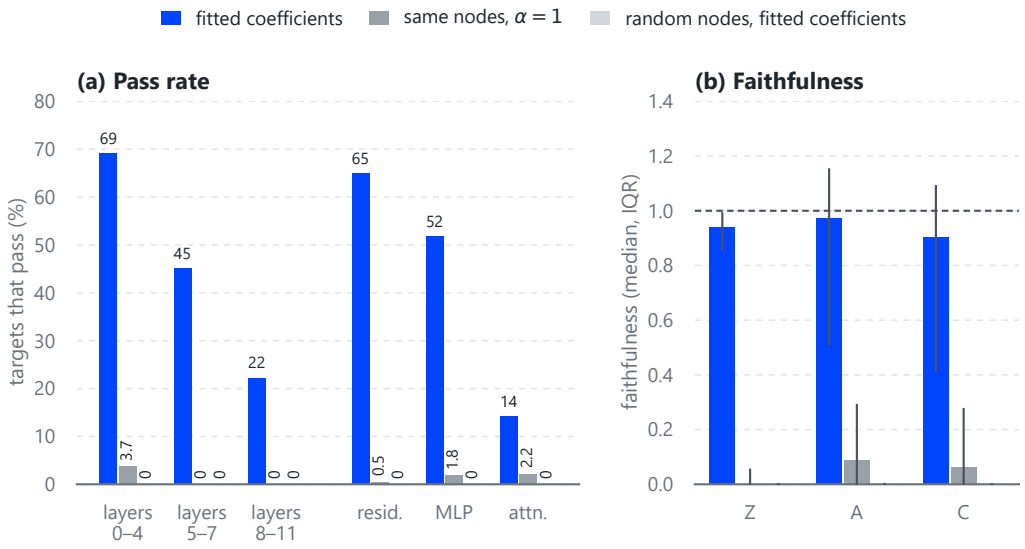



Figure 14: The coefficient triad on the production circuits. **(a)** The share of targets that pass the criterion of Section 4.3, by target depth band and site kind, for the circuits as fitted (blue; the full run, 14,994 circuits), the same nodes with every coefficient set to 1 (grey; 1,746 targets), and random nodes of the same size given the same coefficient fitting (light grey; 747 targets), whose coefficients barely move from initialisation (median $\alpha$ exactly 0.798, the value weight decay alone reaches; Appendix H). **(b)** Median faithfulness under zero (Z), activating-mean (A), and contrast-mean (C) ablation, with interquartile ranges; 1 is perfect (dashed). [*pending: anchor-active and contribution-matched nulls in both panels, DAN-181*]

target's pre-activation, binned by deciles of the production members' own contributions, and excludes the production members: it is a substitution test, asking whether other latents of the same strength can stand in for the selected ones. Each set is fitted exactly as above and scored on held-out and on training contexts, with fitted coefficients and with every coefficient at 1; the latter is the size-matched, nothing-fitted null of prior work (Shi et al., 2024; Seth et al., 2026), and once circuits carry fitted coefficients the fair control fits them on the random members too. Both run on 126 of the sweep targets (14 per depth band and site kind) and on 128 targets per public model, one draw each and a second draw on 32, so that each target's WCM score can also be placed within its own null draws. [*pending: harder nulls: pass rate with interval per null and depth band, median Z / A / C and necessity on held-out and training contexts, the share of coefficients that moved, the WCM score's quantile among each target's draws, the overlap of anchor-active draws with production members, and the contribution-matching shortfall, on all three models, DAN-181*]

**What the coefficients do to members.** Comparing each member's value inside the circuit-only run with its value in the unmodified model, at the target's anchors on held-out contexts, shows that the coefficients do not return members to their natural values (Figure 15a; 600 production circuits, 200 per depth band). Before the coefficients act, the ablated environment leaves 14% of members unchanged, pushes 8–10% out of their site's Top-K, and raises 11–14% to three or more times their natural value; 28–30% lie within 25% of it. After the coefficients fewer do (22–23%), with the spread widening on both sides, and weighting members by their gradient-times-activation contribution to the target gives the same picture. A member's coefficient is unrelated both to how far the ablation moved it and to how much it contributes (rank correlations within ±0.1). What the coefficients change is the target. With every coefficient set to 1, the members alone bring the target to a median 0%, 19%, and 7% of its natural activation under Z, A, and C (21%, 24%, and 12% of its pre-activation), against

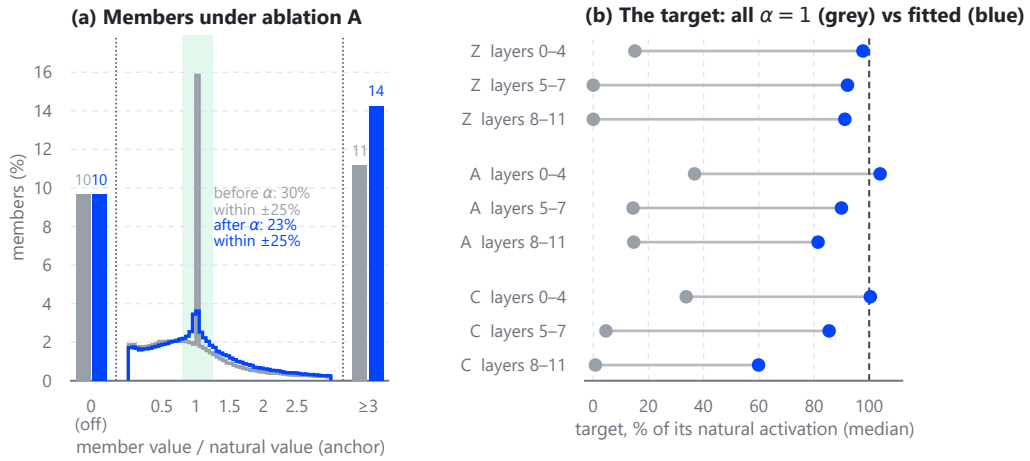



Figure 15: What the coefficients do to circuit members (600 production circuits). **(a)** Each member's value in the circuit-only run divided by its value in the unmodified model, at the target's anchors on held-out contexts, under activating-mean ablation, before (grey) and after (blue) its coefficient; members active in the unmodified model, pooled over circuits. The bars at either side collect members that left their site's Top-K ("0, off") and members at three or more times their natural value; both groups are mostly members that fire weakly in the unmodified model, and together they carry under a tenth of the circuits' contribution to their targets. **(b)** The target's activation in the circuit-only run as a share of its natural activation, median over circuits, with every coefficient set to 1 (grey) and with the fitted coefficients (blue), by ablation method and target depth band.

94%, 97%, and 90% with the fitted coefficients (Figure 15b). To first order the coefficients raise the members' summed contribution only about 1.6×, so most of their effect on the target arises in the layers between the members and the target; we report this as an observation, not a mechanism. Deep and failing circuits also lose members to the ablated environment itself: weighted by contribution, 10% of members of layer 8–11 circuits are pushed out of Top-K under contrast-mean ablation, against 1% at layers 0–4, and 9% in failing circuits against 1% in passing ones. At natural scale ($\alpha = 1$), unweighted circuit masking needs $\sim 10^4$ latents before a deep target passes (Figure 3); we interpret the coefficients as locating most of that cost in redundancy, many latents each carrying a small share of a signal that fewer, rescaled nodes carry alone.

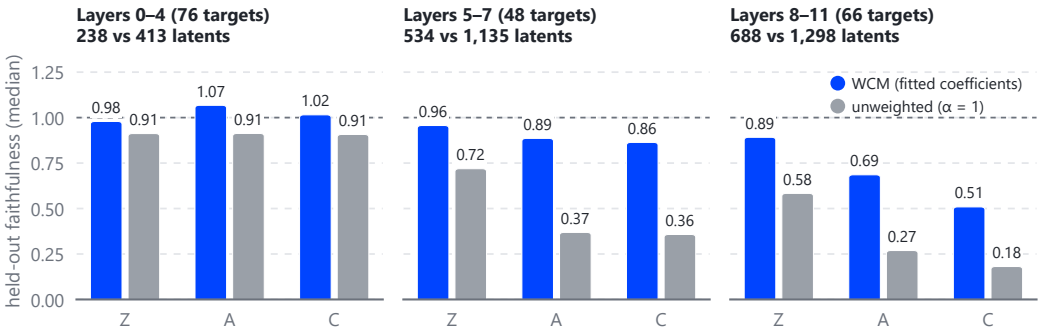



Figure 16: Per-score medians at the production sparsity penalty, $\lambda = 10^{-3}$: held-out faithfulness of weighted (blue) and unweighted (grey) circuit masking under zero ablation (Z) and mean ablation over activating (A) and contrast (C) contexts, by target depth band; panel titles give the median circuit sizes, weighted against unweighted. 1 is perfect (dashed).

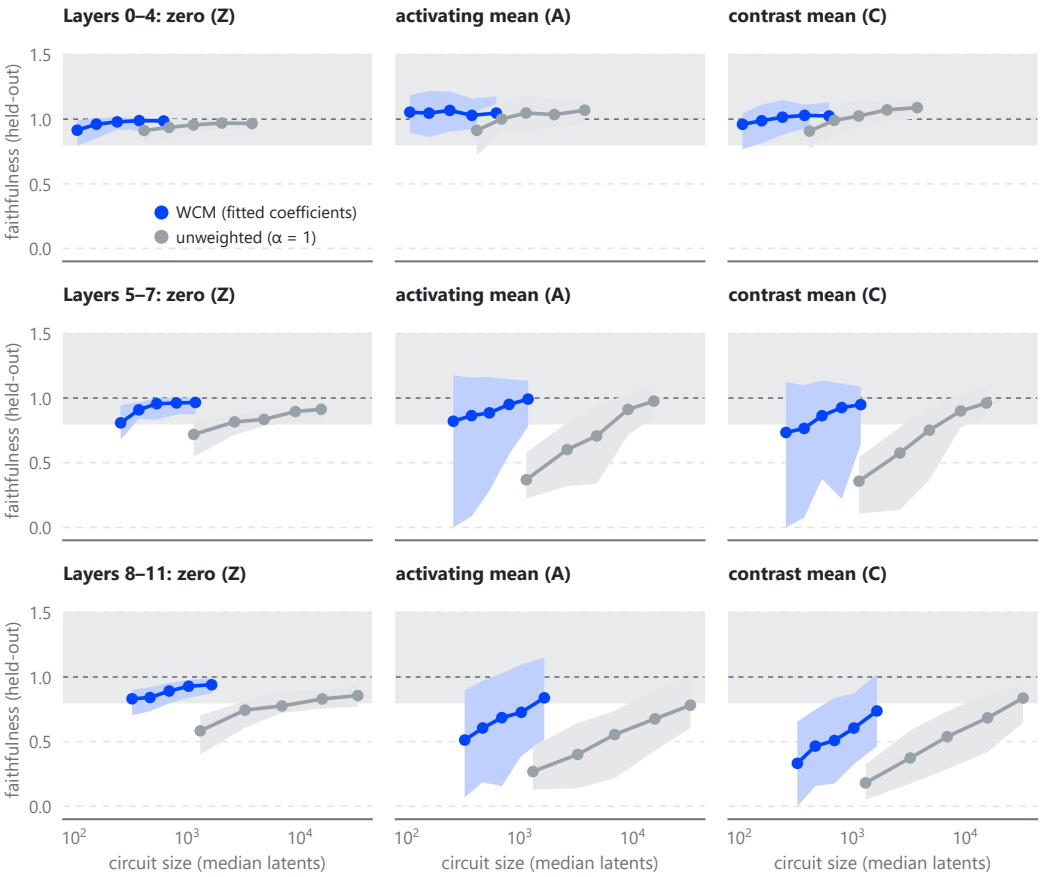



Figure 17: Weighted against unweighted circuit masking by ablation method (rows: target depth band; columns: zero ablation, and mean ablation over activating and over contrast contexts). Median held-out faithfulness and interquartile range against median circuit size, one point per sparsity penalty; the grey band is the pass band $[0.8, 1.5]$. The gap between the methods is smallest under zero ablation, largest under the two mean ablations, and widens with depth.

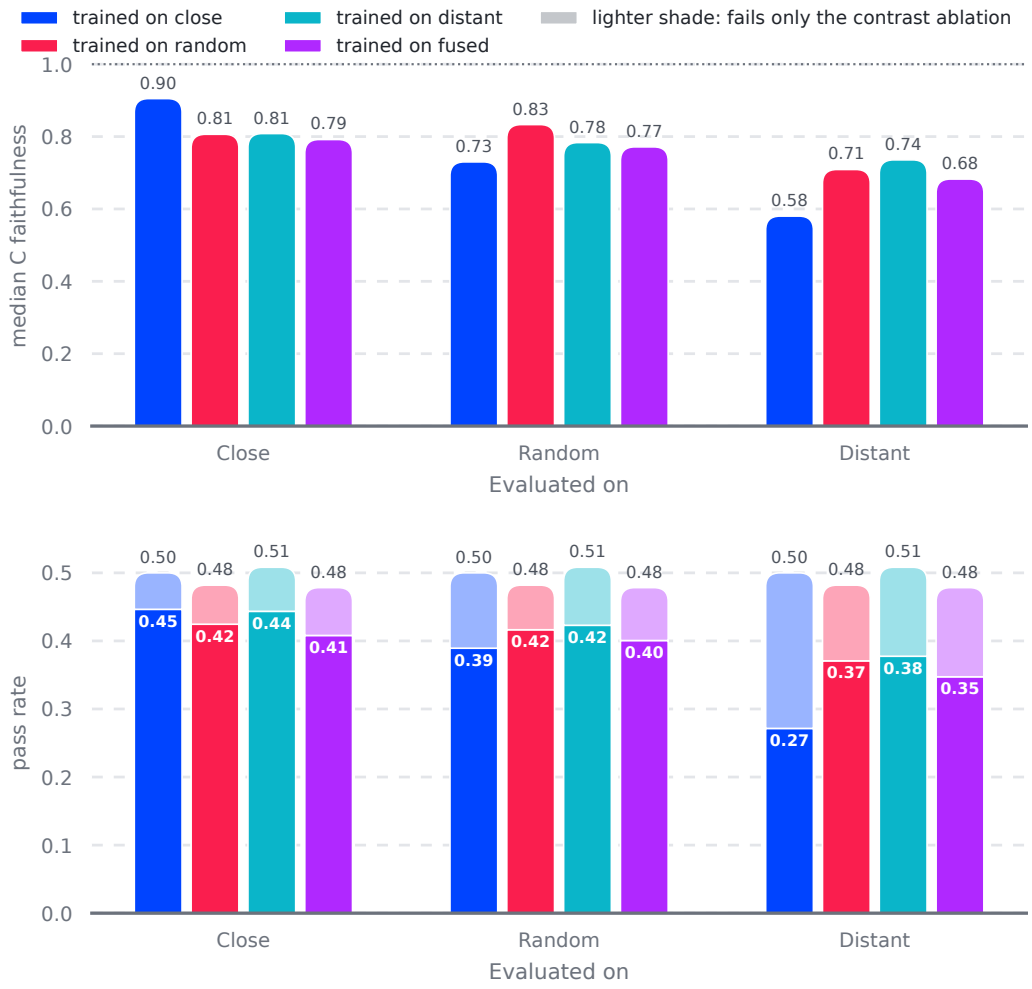



Figure 18: Weighted circuits trained against close contrast contexts generalise worst to other contrast backgrounds. Groups: the contrast set a circuit is evaluated against; bars: the contrast set it was trained against. Top: median faithfulness under the contrast-context mean ablation (C). Bottom: pass rate under the protocol-v1 criterion, each evaluation set supplying its own C; the lighter segment counts circuits that fail only the contrast ablation (zero-ablation and activating-mean faithfulness in band, necessity $\geq 0.9$), and its top, the share that would pass with a perfect C term, is the same for each training source on every background. 280 targets at a matched size of 490 latents; the paired differences are in the text.

# I Weighted Circuit Masking under Each Contrast Source

The C term of WCM trains against the mean over contrast contexts, so the choice of contrast contexts could shape what a circuit learns. We refitted 280 targets of the production run (8 per layer and site kind) under four contrast sources, all checked for silence after Top-K: *close* (the production set, Appendix Q), *random*, *distant*, and *fused*, a third of each. For every target the four sets are disjoint, and every circuit, fitted at $\lambda \in \{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, is scored on the held-out contexts of the close, random and distant sets alike, so the training source is the only difference between arms and all arms are scored on the same contexts. Scores are interpolated to a matched circuit size of 490 latents, the median close-trained circuit at the production sparsity penalty.

Circuits trained against close contexts are the most faithful on close backgrounds and the least on any other (Figure 18). Their median C faithfulness falls from 0.90 on close to 0.58 on distant backgrounds, and on distant backgrounds they pass 0.27, against 0.37 and 0.38 for circuits trained on random or distant contexts (paired differences +0.099 $[0.052, 0.149]$ and +0.106 $[0.065, 0.152]$, 95% bootstrap intervals over targets). Averaged over the three backgrounds, distant-trained circuits pass more often than close-trained ones (+0.046 $[0.014, 0.082]$), random-trained ones by a similar margin that the interval does not resolve (+0.035 $[-0.002, 0.074]$), and random and distant do not differ ($-0.011$ $[-0.045, 0.023]$). The source acts mainly on the C term: zero-ablation faithfulness and necessity are the same across sources (activating-mean faithfulness is somewhat higher for close-trained circuits, median 1.00 against 0.92–0.94), and for each source the share of circuits that pass or fail only under C is the same on every background (0.48 to 0.51), so a different background only moves circuits between passing and failing on C.

**Why close contexts generalise worst.** Variety alone does not explain it. Fused contexts span the widest range of similarity to the target's activating contexts (10th to 90th percentile 0.25 to 0.87, against 0.82 to 0.90 for close, 0.47 to 0.77 for random, and 0.19 to 0.27 for distant), yet fused-trained circuits do no better than random-trained ones ($-0.019$ $[-0.054, 0.017]$). What close contexts have that the others lack is the target's topic: they sit nearest the activating contexts and nearest to firing (median pre-activation 3% of the activating level, against 0 for random and distant). A circuit trained against their mean keeps the topic's features in its background and learns only what separates the target from near misses; against a background without those features it falls short. The same circuits, re-scored with the C mean taken from the nearest, middle or farthest third of each target's random set, show this within a single source: close-trained circuits pass 0.40, 0.40 and 0.33 as the background moves away from the target's topic (nearest minus farthest +0.068 $[0.039, 0.100]$), while random-trained circuits pass 0.43, 0.44 and 0.41 (+0.014 $[-0.007, 0.036]$).

The production run uses close contrast contexts, and the public-model runs keep them so that the model is the only change between models. For new work, random or distant contrast contexts give circuits whose contrast faithfulness depends less on the background they are evaluated against. An earlier 16-target pilot, with the unchecked retrieval store as a fourth source, found that node sets move no more between contrast sources than between two refits with the same source.

# J Specificity

Faithfulness, necessity, and the random-circuit baseline all read the target alone. A circuit can therefore restore its target by lifting the target's whole neighbourhood, its concept, topic, or context type, and pass all three: a concept amplifier rather than a mechanism for that latent. The loss permits it, since reproducing the target's pre-activation constrains the site's stream along one direction of $d_{\text{model}}$, and it leaks into the activation read only when the lifted latents overtake the target in its site's Top-K.

The diagnostic reads the whole code at the target's site, every latent's pre-activation and the SAE's Top-K set at the anchor, in the unmodified model, the empty circuit, and the fitted circuit under each ablation method, on the held-out contexts. It reports the target's rank and Top-K membership; the number of latents lifted above the unmodified model's Top-K cut, against the empty circuit; the overlap of the circuit's Top-K set with the unmodified one; and the faithfulness of *siblings*, latents in the unmodified Top-K at the anchor on at least half of the target's strongest training contexts, computed exactly as the target's, against a control group of the site's other active latents. A circuit is flagged as an amplifier when, under at least one ablation method and on the pre-activation read, its target is restored to at least 0.5 and its median sibling to at least 80% of the target's faithfulness.

The rank-keep term (Section 4.2) keeps the target in its Top-K more often but does not reliably remove amplifiers, so specificity is reported for every circuit rather than claimed for the method. Most circuits are specific (15,004 circuits): under Z, A, and C the target is restored to a median 0.94, 0.99, and 0.92 of its pre-activation, its siblings to 0.26, 0.46, and 0.41, and the site's other active latents to 0.14, 0.28, and 0.24. Under the two mean ablations, circuits lift a median of 578 and 517 latents

above the unmodified Top-K cut, against 5 and 4 for the empty circuit, and the target stays in its Top-K on 84% and 80% of held-out anchors. 30% of circuits are flagged as amplifiers under at least one ablation method (44% of MLP, 30% of residual-stream, and 11% of attention-output targets), including 38% of passing circuits; 29% of all circuits pass and carry no flag.

The loss terms we tried for rank and specificity before adopting rank-keep are in Appendix T.1.

# K Resample Ablation and Reads at Every Position

All three of our ablation methods replace a removed latent with zero or a mean. Causal scrubbing (Chan et al., 2022) sets a stricter test: everything outside the circuit takes its value on another real context. We scored the production circuits of the 747 targets that have a random-circuit baseline this way. At each position a removed latent takes a donor context's sparse code (so the donor's sparsity is kept), with donors drawn from the target's activating training contexts (A), its contrast contexts (C), or random corpus text (R); the scores and the pass rule are otherwise unchanged. With A and C donors, 0.1% of circuits pass, against 43.1% under the production test, with median faithfulness 0.18 (A), 0.04 (C), and 0.01 (R); random circuits pass on none.

The failure is one of closure, not of membership, for residual-stream targets. In a 45-target diagnostic with C donors, holding the members at their own values in the base context restores residual-stream targets to a median 0.92 (MLP 0.39, attention 0.11), while members recomputed from the mixed stream do not; the node set carries the signal, but the members lose the inputs they receive from outside the circuit. The coefficients show the same dependence: with the base context as its own donor, so that nothing is ablated, residual-stream circuits overshoot to a median 2.16, because coefficients fitted to stand in for an ablated complement over-drive the target when the complement is present.

Nothing in training asks a circuit to survive a real background, so we added one: a fourth ablation method whose replacement values are donor codes, drawn afresh at every step from contrast and random corpus contexts disjoint from the evaluation donors. In a 27-target pilot, paired against the production configuration refitted on the same hardware, it raises the resample pass rate from 0% to 33% and the production pass rate from 37% to 52%, at 1.15× the circuit size. Its side effect is that the target now also fires at nearby positions of the activating passage where it is naturally silent; a penalty on lifts at those positions roughly halves the effect, and with both terms at weight 0.25 the resample pass rate is 37%, the production pass rate 59%, and the size 1.35×. With that configuration fixed in advance, 99 further targets (11 per depth band and site kind) confirm the pilot: the resample pass rate rises from 0% to 30% (39%, 36%, and 15% by depth band), the production pass rate from 44% to 57%, and the size by 1.22×, while the correlation with the target at every position stays where it was (0.77 against 0.79). Deep targets gain least on the resample test but most on the production test (21% to 45%).

Every score above, like the pass rule, reads the target at the anchor. Read instead at every position of 16 held-out activating and 16 held-out contrast contexts, the circuits track the target only in part. Under zero ablation the median correlation with the target's activation is 0.59 (0.75, 0.58, and 0.50 by depth band), and beyond layer 4 the variance explained is negative ($-1.06$ at layers 5–7, $-2.14$ at layers 8–11). The cause is firing where the target is silent: on 31% and 74% of contrast-context positions in those bands, against 1% at layers 0–4. Under the contrast-mean ablation the reads are closer (median correlation 0.66; 3% of contrast-context positions), and passing circuits track the target far better than failing ones (correlation 0.88 against 0.35; variance explained 0.46 against $-0.25$). A circuit is thus a mechanism for the target's activation where the target fires, not a simulation of the target at every position, and deep circuits under zero ablation overfire away from the anchor.

Faithfulness is therefore relative to the ablation methods a circuit is trained and tested against, and robustness to a stricter test can be trained for by giving training the backgrounds the circuit must survive. Every result in the main text uses the three methods of protocol v1.

| Target | What it computes | Minimal pair (clean model) | Story groups removed: Z / A / C (random) | Caveat |
|---|---|---|---|---|
| 9.resid.20419 (Section 5.3) | a date or place read as classical Greek antiquity | "In the 5th century BC, the philosopher . . . " 76%; "In the 19th century, . . . " 0% | 0.20/0.20/0.19 (0.49/0.98/0.74) | "BC" is partly lexical ("The BC Lions" 44%) |
| 10.resid.15497 | the possessive after a name that carries a law, before "law" appears | "Snell's" 94%, "Ohm's" 80%; "Darwin's", "Picasso's" 0%; invented "Zorbel's" 76% in electromagnetism, 0% in painting | 0.00/−0.12/0.03 (0.54/0.92/0.64) | amplifier: lifts ∼7,900 latents, target rank 1 → 4 |
| 8.resid.31095 | "but also" when a sentence attributes causes | "influenced not only by memory but also" 89%; "involves not only memory but also" 0% | 0.19/0.41/0.28 (0.73/0.93/0.81) | one latent alone gives most of the drop |
| 8.attn.29991 | acid–base dissociation | acid–base sentences 46–87%; other chemistry 0–16% | 0.25/0.11/0.16 (0.61/1.19/0.80) | "Her acid wit" 38% |
| 9.resid.7640 | the pivot in "not X, but rather Y" | negated clause, then pivot 37–124%; same comma without negation 0–13% | 0.00/0.03/0.00 (0.42/0.80/0.39) | fragile: random removals also hurt; lifts 15–21k latents |
| 9.resid.17596 | the second "as" of a comparison about importance | "as important for health as" 79%; "as fast as" 0% | importance group no better than random | not validated; lifts ∼30k latents |

Table 5: Six case studies from the hunt. Knock-outs remove the case's named story groups together and report faithfulness under zero (Z), activating-mean (A), and contrast-mean (C) ablation; in brackets, the mean over 8 draws of the same number of random members of similar contribution rank. In every validated case, groups named as controls (generic or lexical members) are no worse than random.

# L Case Studies

**How the cases were found.** We read passing circuits for targets at layer 6 or deeper (2,238 circuits) with tools that list, for each circuit, its members ranked by contribution at the target's anchor ($\alpha$ times the member's mean activation there times its decoder norm), each member's own top contexts and peak tokens, its engagement on the target's activating against contrast contexts, and its first-order logit effect, and that run hand-written sentences through the clean model. Candidates were kept when a minimal pair of sentences separated the target cleanly. Group names come from the members' own contexts and logit effects, not from automatic interpretation.

**Validation.** Each case is tested three ways. *Knock-outs*: the circuit is re-scored with one named group removed (its members ablated like any non-member), on the held-out strongest contexts with the production scorer, against the same number of members of similar contribution rank (±15 ranks, excluding every named member and a generic core of latents that fire on most text), over 8 random draws. *Probes*: the target's activation in the clean model on hand-written sentences, as a percentage of its mean activation on its held-out strongest contexts. *Specificity*: the diagnostic of Appendix J. Table 5 summarises six cases; the remaining tables are in the repository.

**More on the antiquity circuit.** Two further measurements qualify Section 5.3. First, a generic core of seven layer-0 and layer-1 latents that fire on most text is unnecessary under the mean ablations

but essential under zero ablation (removing it takes Z to 0.00, A 0.99, C 0.88): with everything else zeroed, these latents keep the stream within the range the later layers expect. Second, run alone on the template sentences under contrast-mean ablation, the circuit keeps ancient Greece and Athens (27) and Rome (18) well above other places (6–8, from Egypt to medieval France; the empty circuit gives 0) but separates dates only weakly (5th century BC 6.9, 19th century 3.8): it was fitted on passages about classical Greece, and its date route is the weaker one, as the knock-outs also show.

**What the hunt did not find.** No latent we examined tracked whether a stated fact was correct: knowledge appears as latents for entities, eras, and domains, and in the sentences we tried, a wrong date or a false definition activated them as much as the true one, or more.

# M Circuits for Randomly Drawn Targets

The case studies were chosen; the circuits in Table 6 were not. We drew one target uniformly at random from each depth band and site kind of the production run (seed 20261012, non-vacuous circuits, any outcome) and list each circuit's scores and its three largest members by contribution share at the target's anchor, excluding generic hub latents. Four of the nine pass, close to the run's 46%. The labels are hypotheses written by a language model from each latent's own strongest contexts and logit effect, not verified readings: the target labels come from the run-wide circuit descriptions, and each member was labelled from its own evidence alone.

The shallow token detector `0.mlp.12557` ("control") rests largely on two members that detect the same token, with 8.5% and 7.5% of the contribution, while the deep circuits spread over many members with 1–4% each that carry topic and construction (`8.mlp.2453`, `9.resid.10786`). The failing attention target `5.attn.10168` is near-threshold and reads 0 under every ablation method, as Appendix G describes. Five of the six circuits at layers 5–11 are flagged as amplifiers, against 30% over the run (Appendix J); nine draws are too few to read this as a trend with depth.

| Target | Target label | Size | Z / A / C | Nec. | Ind. | Outcome | Top three members (label) |
|---|---|---|---|---|---|---|---|
| 3.attn.9767 | Token following "Capital" in finance terms | 257 | 0.97 / 1.10 / 1.16 | 1.00 | 0.58 | pass | 1.resid.18903: Word-start fragments in Title Case (unclear)<br>0.resid.4768: Title-case word-start fragment (unclear)<br>0.mlp.4768: Title-case word-start fragments (unclear) |
| 0.mlp.12557 | Control word detector | 26 | 0.88 / 0.95 / 0.99 | 0.72 | 0.39 | fail | 0.attn.16913: Token "control"<br>0.attn.5256: Token "Control"<br>0.attn.19725: Title-case word starts (unclear) |
| 0.resid.27712 | Cell division terms (chrom, pol, II) | 102 | 0.86 / 0.66 / 0.81 | 1.00 | 1.24 | fail | 0.mlp.205: 'chrom' word-start token<br>0.mlp.4393: "fect" in "Perfect" tense headings<br>0.mlp.21122: Arabic mid-word letters (unclear) |
| 5.attn.10168 | 'men' in "menstrual" | 274 | 0.00 / 0.00 / 0.00 | 1.00 | 0.00 | fail$^{t}$ | 4.resid.19412: Mid-word fragments, unclear<br>3.resid.12927: Word after "the" (unclear)<br>3.resid.18667: Mixed tokens, unclear (further/later) |
| 5.mlp.15947 | "not limited to" negated limitation | 543 | 0.99 / 1.19 / 1.19 | 1.00 | 0.76 | pass$^{a}$ | 4.resid.21357: Token "limited"<br>4.resid.25898: "not limited to" phrase<br>4.resid.26922: Negated verb "do not contain/have" |
| 7.resid.29770 | food token detector | 721 | 0.96 / 0.87 / 1.27 | 1.00 | 1.24 | pass$^{a}$ | 6.resid.32006: "red" in ingredient (food prep)<br>5.resid.3962: Grill/barbecue word fragments (food)<br>5.resid.17590: Token "food" before noun |
| 9.attn.38891 | Connectors in lists of consequences | 613 | 0.88 / – / 0.11 | 1.00 | 1.21 | fail$^{a}$ | 8.resid.33386: Predicts negative-consequence words<br>7.resid.39521: Subject before past-tense verb<br>7.attn.15390: Humanism/Renaissance context function words |
| 8.mlp.2453 | Structure nouns followed by that/can | 870 | 0.92 / 0.68 / 0.41 | 1.00 | 1.46 | fail$^{a}$ | 8.attn.4808: Engineering-context function words (unclear)<br>7.attn.27777: Composite materials engineering context (unclear)<br>7.attn.18401: Plural subject, predicts plural verb |
| 9.resid.10786 | "global" as modifier | 920 | 0.92 / 1.01 / 1.08 | 1.00 | 1.30 | pass$^{a}$ | 8.resid.30012: Capitalized "Global" at title start<br>8.resid.26956: Growing awareness → environment<br>7.resid.11194: Pre-"international/global" context words |

Table 6: Circuits for nine randomly drawn production targets, one per depth band (rows grouped L0–4, L5–7, L8–11) and site kind, seed 20261012. Size is the number of upstream latents; Z, A, C are faithfulness under the three ablation methods, Nec. necessity, and Ind. sufficiency to induce with coefficients, all on held-out contexts; "–" is an undefined score, which counts as a fail. $^{a}$ flagged as an amplifier; $^{t}$ near-threshold target. Labels are hypotheses written by Claude Sonnet 5.5 from each latent's own strongest contexts and logit effect; members are ranked by contribution share at the target's anchor, generic hub latents excluded.

# N Extended Related Work

**Discovery methods and their settings.** Each family of discovery methods is built for its own setting. ACDC (Conmy et al., 2023) removes edges of a model's computational graph one at a time while an output metric holds, and Edge Pruning (Bhaskar et al., 2024) optimises an edge mask by gradient descent so that the pruned model stays faithful to the full model's predictions; both work over components such as heads and MLPs, and both are anchored at an output rather than a latent. Attribution patching (Nanda, 2023; Syed et al., 2023) replaces each activation patch with a first-order estimate, AtP* (Kramár et al., 2024) adds two changes that address the two classes of failure behind its false negatives, and EAP-IG (Hanna et al., 2024) scores edges by integrating gradients along a path from corrupted to clean input embeddings. Our path-attribution baseline adapts this estimator to nodes, integrating jointly over the upstream SAE sites from their mean-ablation values (Appendix C). Transcoders (Dunefsky et al., 2024) replace MLPs with sparse, wider layers whose circuits factorise into input-dependent and input-invariant terms, attribution graphs (Ameisen et al., 2025; Lindsey et al., 2025) build one graph per prompt over cross-layer transcoder features, and PIE (Chen et al.,

2026a) prunes cross-layer transcoder features by their gradient-weighted write contributions before interpreting them. The transcoder-based methods need a substrate our SAE bank does not provide, and produce a graph per prompt where we fit one circuit over many contexts, so we do not run them here.

**Masks, scalars, and ablations.** Learned masks for circuit discovery descend from subnetwork probing (Cao et al., 2021), applied to computational-graph vertices by Conmy et al. (2023) and to edges by Edge Pruning (Bhaskar et al., 2024) and by uniform gradient sampling with optimal ablation (Li and Janson, 2024), on $L_0$-gate and continuous-sparsification machinery (Louizos et al., 2018; Savarese et al., 2020). Masks over SAE latents have been learned to localise attributes (Chaudhary and Geiger, 2024), a procedure adopted by benchmarks (Mueller et al., 2025; Karvonen et al., 2025), and to find behavioural circuits by continuous sparsification (Caples et al., 2025); with its coefficients fixed at 1, its objective moved to a single latent, and one mask shared across positions, the latter is our unweighted circuit masking. Every such mask keeps a node at its natural scale or ablates it. The continuous coefficients of Li and Janson (2024) interpolate between the two and play the role of our gates; what they learn besides is a constant for each *removed* component, the dual of our scale for each kept one. That constant minimises the ablated model's expected loss, so a component's importance is what it carries beyond the best constant; we keep the three ablations fixed and defined by data (zero, and the means over activating and over contrast contexts), so that what absence means is set before training rather than fitted with the circuit.

Learned per-unit scales appear with other jobs: scales that double as the pruning criterion (Liu et al., 2017; Huang and Wang, 2018; You et al., 2019); scales for the units kept after pruning, fitted after the mask is fixed (Kwon et al., 2022; He et al., 2017); steering multipliers (Stoehr et al., 2024); sparse fitted weights on SAE latents for a task vector (Kharlapenko et al., 2025); regression weights on feature-to-feature edges (Yin et al., 2026); and manual rescaling of the nodes of a pruned circuit (Gao et al., 2025). Gate-times-magnitude parametrisations separate membership from strength in sparse training and fine-tuning (Louizos et al., 2018; Guo et al., 2021; Lai et al., 2025), where a closed gate means no change or zero rather than an ablated node. Feature-preserving subnetworks have also been pruned by saliency in vision models (Hamblin et al., 2022).

Circuits are routinely evaluated under several ablations (Miller et al., 2024; Mueller et al., 2025) but discovered under one. The nearest objectives sum the clean-to-corrupted and corrupted-to-clean directions of one counterfactual (Chen et al., 2025), or certify a circuit against every patch value in a bounded set (Hadad et al., 2026). Training under several ablations is the circuit-level counterpart of attribution averaged over several baselines (Kapishnikov et al., 2019; Erion et al., 2021); it is not resample ablation, which our circuits mostly fail (Appendix K).

**Nulls, variance, and non-uniqueness.** Random circuits as a null follow Shi et al. (2024), who test whether a candidate is more faithful than about a hundred random circuits of at least its size, sampled as paths through the model, and CircuitKIT (Seth et al., 2026), which reports a $z$-score against random subgraphs of equal size. In both, the random circuits are sampled and ablated but not fitted, which matches circuits that are not fitted either. Our circuits fit a coefficient per node, and fitting alone might rescale an arbitrary set into reproducing a target, so our random circuits receive the same coefficient fitting before scoring (Section 4.3); with every coefficient at 1 they reduce to the nothing-fitted null. The broader concern is the one Méloux et al. (2025a) raise: interpretability methods produce plausible explanations even for randomly initialised networks, so findings are better treated as statistical estimates tested against alternative hypotheses. Causal scrubbing (Chan et al., 2022) is a stricter test in this spirit, resampling everything outside the hypothesis from other inputs (Appendix K).

Circuit discovery is also variable. EAP-IG circuits vary with the data batch, the prompt template, and the example (Wu et al., 2026); on one task, defensible analytic choices change the derived claim for most pairs of specifications (Mahale, 2026); and Certified Circuits stabilise membership by randomised subsampling of the data, keeping only components included consistently (Anani et al., 2026). On small networks where every circuit can be enumerated, many circuits reproduce the same behaviour exactly (Méloux et al., 2025b), and in language models structurally distinct, low-overlap

circuits support the same task (Chen et al., 2026b), with circuits for one task differing across its inputs (Makou et al., 2026). Our node sets are non-identifiable in the same way, which is why a circuit's function, not its exact member list, is what we report as the finding (Appendix O).

**Self-repair and inhibition.** Ablation-based tests meet compensation: when one attention layer is ablated, later layers make up part of its effect (McGrath et al., 2023); the self-repair of single heads persists over the full training distribution, partial and variable across prompts, carried in part by the final LayerNorm scale and by sparse anti-erasure neurons (Rushing and Nanda, 2024); and copy-suppression heads, which lower the logits of tokens that earlier layers would copy from context, account for part of it (McDougall et al., 2024). Conditional co-ablation finds backup components that become important only once a primary set is removed (Gong et al., 2026). Our scores intervene on the circuit as one set (Section 4.3), so they measure what the set does with any compensation inside the model included; per-member roles, where backups matter most, are not claimed.

**Position-aware circuits.** Position handling divides prior work into three strategies, each right for its setting: position-indexed nodes on templatic data and scores summed over positions otherwise (Marks et al., 2025), per-instance graphs built one prompt at a time (Ameisen et al., 2025), and schema-aligned positions across variable-length examples (Haklay et al., 2025). WCM needs none of these commitments: one mask is shared across positions and nodes are recomputed at every position, which suits the setting none of the three targets, one reusable circuit over arbitrary non-templatic text; our attribution baselines instead select per position and take the union over the causal prefix (Appendix C.2).

**Contrast contexts.** Retrieved non-activating examples appear in automated interpretability for explanation scoring (Paulo et al., 2024); Paulo and Belrose (2025a) find evaluations collapse when negatives come from neighbouring features, convergent with our finding that close contrast contexts generalise worst to other backgrounds (Appendix I); and O'Neill and Bui (2024) design corrupted negatives for task-level circuit identification. We use retrieved contrast contexts for causal discovery rather than explanation scoring, and treat their construction as a controlled variable with measured consequences.

**The SAE basis.** Static structure among latents is well documented: co-occurrence and decoder geometry reveal modularity (Li et al., 2025), coactivation-derived components steer behaviour (Deng et al., 2026), meta-decompositions show latents are not atomic (Leask et al., 2025), cross-layer feature-flow graphs track latents through the forward pass (Laptev et al., 2025), though decoder similarity misses most causally strong feature interactions (Droste et al., 2026), and Jacobian SAEs train adjacent dictionaries for sparse computation between them (Farnik et al., 2025). But that structure is correlational, geometric, or weight-based, where the relationships discovered here are context-conditioned and intervention-tested. Weight-based connections need that test: virtual weights between interpretable components include *interference weights*, large connections between components that never interact on the data distribution (Turner et al., 2026). Intervention-based latent graphs have been mapped at scale in single-cell foundation models, by ablating one SAE feature at a time and measuring the change in downstream features, finding inhibitory edges dominant (Kendiukhov, 2026a) and a heavy-tailed hub structure with redundancy that grows with interaction order (Kendiukhov, 2026b); those graphs measure the effect of one feature at a time, where our circuits are fitted jointly for each target and tested as a set. The basis itself is imperfect: reconstruction errors are pathological (Gurnee, 2024), although much of the error is linearly predictable from the activation (Engels et al., 2025); SAEs trained from different seeds learn largely different features (Paulo and Belrose, 2025b); downstream utility is mixed (Smith et al., 2025); and units are not canonical (Leask et al., 2025). Interventions on the basis are imperfect too: a behaviour suppressed by clamping SAE features can be recovered through the reconstruction residual while the features stay clamped (Cui et al., 2026); the effect of ablating single-token features on their token's logit differs more across SAE families than across model scales (Cho et al., 2026); and dictionaries trained only on what an affine map from an earlier

layer cannot explain recover more of the loss when several layers are replaced at once (Poduval et al., 2026). This conditions our claims: reconstruction residuals stay as explicit anchors, evaluation intervenes on measured activations, and the SAE basis is treated as a useful but imperfect sample of the model's internal variables.

# O Node Sets Are Non-Identifiable; Function Is Not

The masking engine is bit-deterministic at fixed settings on one GPU type (two identical runs agree exactly, with *zero* node flips), so small perturbations are controlled experiments on the objective, and each one we tried gave the same verdict. In pre-v1 probes, five on unweighted circuit masking and two on WCM, each perturbation left faithfulness roughly unchanged but exchanged nodes. A float32 and a bfloat16 run differ only in rounding, yet exchange 16.5% of nodes at layer 8 (Jaccard 0.835) at unchanged faithfulness and size. Strengthening the sparsity penalty swaps nodes rather than nesting them (containment 0.49–0.80 between consecutive steps); no gate ever freezes, so the stopping step is another sampling axis; 8–25% of nodes sit within 0.05 of the decision cut and are load-bearing; and short-budget and long-budget runs at matched size share only 0.40–0.53 of their nodes. Recomputing the mean-ablation values without a quarter of their contexts, a change that leaves the pass rate and median faithfulness unchanged, moves node sets as far as a change of hardware (median Jaccard 0.66). Under protocol v1 the effect holds at scale: the same settings on a different GPU type give node sets with a median Jaccard of 0.60 (0.32–0.96 over 32 targets), and a refit with reordered training contexts shares a median 64% of the original circuit's latents (below). A discovered node set is one sample from an equivalence class of near-optimal solutions to (P), so we report node sets as equivalence classes and let set-level and functional claims carry the conclusions.

**A stable core at scale.** To measure this on the production run, we refitted 175 of its targets (5 per layer and site kind) under protocol v1 with one change: the training contexts visited in a different fixed order. The refit ran on a different GPU from the production fit, and a refit is bitwise reproducible on the same GPU type but not across types, so the two fits differ by the reordering and by float-level differences in the hardware. The *core* of a target is the set of latents in both circuits; its coefficients are refitted on the core alone, as for the random baseline, and it is scored with the production scorer. Table 7 gives the result. The two fits share a median 64% of the original circuit's latents (Jaccard 0.54), less at depth. The second fit is smaller (median 402 against 492 latents; smaller for 160 of the 175 targets) but passes about as often as the first (40% against 43%), and where both fits pass, the core alone passes on 48 of 56 targets, and its median necessity is 1.0 in every depth band. Which latents a circuit contains is therefore partly an accident of optimisation, while a core of about two thirds of it carries the function.

| | | | **median size** | | | **pass rate (%)** | | |
|---|---|---|---|---|---|---|---|---|
| **Layers** | **targets** | **Jaccard** | **fit 1** | **fit 2** | **core** | **fit 1** | **fit 2** | **core alone** |
| 0–4 | 70 | 0.60 | 241 | 201 | 172 | 67.1 | 62.9 | 55.7 |
| 5–7 | 45 | 0.51 | 592 | 499 | 346 | 35.6 | 28.9 | 26.7 |
| 8–11 | 60 | 0.45 | 736 | 593 | 405 | 21.7 | 21.7 | 11.7 |
| all | 175 | 0.54 | 492 | 402 | 297 | 43.4 | 40.0 | 33.1 |

Table 7: Stability of circuit membership under a refit. Fit 1 is the production circuit; fit 2 refits the same target with the training contexts in a different order, on a different GPU; the core is the latents of both, with coefficients refitted on the core alone. Jaccard and sizes are medians over targets; pass rates use the protocol-v1 criterion on held-out strongest contexts.

**From circuits to chains.** A weighted circuit is one hop; because every circuit is indexed by its target, circuits can in principle be composed by splicing the circuit for an upstream latent into the

circuits that use it and re-validating by intervening at the earliest nodes. Non-identifiability sets the boundary condition: each link is one sample from an equivalence class, so composition must manage classes of node sets, not single ones.

# P Pseudocode

Algorithm 1 summarises weighted circuit masking (Section 4.2) as run in the production run.

**Algorithm 1** Weighted circuit masking, WCM[Z+C+A] with rank-keep (one target $f$; protocol v1, Table 3)

**Require:** target $f$; activating contexts $\mathcal{P}_f$ (train on 32 strongest + 16 mid-band; hold out 16 + 16) and contrast contexts $\mathcal{N}_f$ (48 train / 16 held out); $\lambda = 10^{-3}$; $\gamma_{\mathrm{C}} = \gamma_{\mathrm{A}} = 0.25$; $\gamma_{\mathrm{R}} = 3 \times 10^{-3}$; steps $S = 400$

target $\leftarrow$ natural pre-activation of $f$ at the training anchors (no-grad forward); $\nu \leftarrow \mathrm{mean}(\mathrm{target}^2)$

$r \leftarrow \min(K,$ rank of $f$ among its site's other latents) at each training anchor, unmodified model

ablation values: C $\leftarrow$ dense site means over the *training* contrast contexts; A $\leftarrow$ over the *training* activating contexts

initialise $\theta_s = 4$, $\psi_s = \mathrm{softplus}^{-1}(1)$ for every upstream site $s$

**for** step $= 0, \ldots, S-1$ **do**

  $T \leftarrow 0.05^{\,\mathrm{step}/(S-1)}$; gates $m_s \leftarrow \sigma(\theta_s/T)$; coefficients $\alpha_s \leftarrow \mathrm{softplus}(\psi_s)$ ▷ geometric anneal $1 \to 0.05$

  **for** $\pi \in \{\mathrm{Z}, \mathrm{C}, \mathrm{A}\}$ **do** ▷ same $\theta, \psi$ under all three ablation methods

    forward with kept latents at $\alpha\times$ live value, closed latents at their $\pi$ ablation value

    $\mathcal{L}_\pi \leftarrow$ MSE of $f$'s pre-activation against target; $\mathcal{R}_\pi \leftarrow$ mean $\mathrm{relu}(\tau_\pi^{(r)} - \hat{a}_f)^2$

  **end for**

  AdamW step on $\sum_\pi \gamma_\pi(\mathcal{L}_\pi + \gamma_{\mathrm{R}}\mathcal{R}_\pi)/\nu + \lambda \sum_{s,i}\left[\sigma(\theta_{s,i}) + (1 - \sigma(\theta_{s,i}))\,|\alpha_{s,i} - 1|\right]$ ▷ penalties use $\sigma(\theta)$, not $\sigma(\theta/T)$

**end for**

circuit $\leftarrow \{\,(i, \alpha_i) : m_i > \frac{1}{2}\,\}$; evaluate on the held-out contexts under Section 4.3

**return** circuit

# Q Context Stores and Contrast Contexts

## Q.1 Context Store Formalism

The collection pass assigns each latent a scored context set $C_f = \{((x,t), a_f(x,t)) : a_f(x,t) > 0\}$. The top-context store retains the $K_{\mathrm{top}}$ highest-scoring active contexts,

$$\mathrm{TopCtx}_K(f) = \operatorname*{arg\,top}_{S \subseteq C_f,\ |S| = K_{\mathrm{top}}} \sum_{((x,t),s) \in S} s,$$

approximating the upper tail of the conditional activation distribution. The mid-context store is a bounded reservoir sample from the sigma band,

$$\mathrm{MidCtx}_M(f) \sim \mathrm{ReservoirSample}_M\big(\{(x,t) : \mu_f + 0.5\,\sigma_f \le a_f(x,t) \le \mu_f + 1.5\,\sigma_f\}\big),$$

maintained without sorting every active occurrence. Mid contexts distinguish a latent's ordinary operating regime from its most vivid examples and provide probe data where extreme contexts would be too narrow.

**Contrast-context retrieval.** Each sequence $x$ is represented by $r(x) \in \mathbb{R}^{1024}$, the final layer's residual stream mean-pooled over positions, captured once during collection. The attribution baselines use a retrieval store: for each target, the 64 sequences most similar, by exact cosine search, to the normalised centroid of its activating contexts, excluding those contexts, with no check for silence. Its index samples $2 \times 10^5$ sequences, so targets with no activating context in the sample fall back to one shared default set (38% of TuringLLM targets).

**Contrast contexts for WCM.** Weighted circuit masking and its evaluation instead select contrast contexts per target at fit time. Each of the target's strongest stored contexts $\mathcal{P}_f^{\text{top}}$ is represented, read from the index when present and otherwise computed by a forward pass; candidates from the sampled corpus are ranked by their highest cosine similarity to any of them, $\max_{x_p \in \mathcal{P}_f^{\text{top}}} \langle r(x_p), r(x) \rangle$, with all of the target's stored activating contexts (strongest and mid-band) excluded; the 8,192 most similar are then run through the model in rank order, and each is kept only if the target is outside its site's Top-K at every position, until 64 are kept. A target with fewer than 4 kept is skipped (19 TuringLLM targets), and one with 4 to 63 uses what it has (10 targets). The silence check rejects between 14% and 51% of the closest candidates on the targets we inspected, so unchecked retrieval would place many contrast contexts where the target fires. The same selector with random or maximally dissimilar ranking gives the *random* and *distant* sources of Appendix I.

**Logit faithfulness.** The output-level score of Appendix E is the faithfulness formula of Section 4.3 with the target's value replaced by the output metric $m$, the mean log-probability of the actual next token at the anchor: $(m(C) - m(\varnothing))/(m(M) - m(\varnothing))$, with every in-scope latent outside the circuit mean-ablated.

# R Evaluation Protocol Studies

## R.1 Sparsity-Preserving Mean Ablation

The classic dense mean ablation assigns every latent outside the circuit its mean, producing a stream with tens of thousands of simultaneously active latents per position, outside the state space a Top-K SAE can produce. The sparsity-preserving variant applies the means and then re-imposes the per-position Top-K (kept latents, then the largest means of the other latents up to $k$, rest zero), which restores the keep-all identity to exactly 1.0. At scale (32 targets, paired) the sparsity-preserving variant does not systematically shift faithfulness: it raises both $a_C$ and $a_\varnothing$, leaving the ratio a wash (50W/46L). It therefore functions as a cleaner protocol rather than a score inflator; but at small node sets and tight Top-K budgets the two variants can disagree on *rankings*, which is why the body's comparisons name their ablation method.

## R.2 Sources of the Mean Ablation

The empty-circuit baseline $a_\varnothing$ is a measured object with its own failure modes. Means over *activating* contexts leak target signal into $a_\varnothing$ increasingly with depth ($a_\varnothing/a_{\text{pos}} = 0/23/30/26\%$ at layers 2/8/9/10); means over the target's *contrast* contexts hold $a_\varnothing = 0.0000$ on every target measured, even under the hardest close contrast contexts. These pre-v1 measurements, on four targets, are the source of the leakage figure in Section 4.3.

# S Dense ReLU Dictionaries

The method is scoped to Top-K dictionaries because of a pre-v1 test on Pythia-70m with the dense ReLU dictionaries of Marks et al. (2025): six residual-stream targets at layers 2 and 4, an earlier form of weighted circuit masking, and held-out contexts. The method replicated: 11 of 12 fits (six targets at two sparsity penalties) were faithful under zero and mean ablation (0.80–1.00) and necessary

(0.81–1.0) on held-out contexts. The random-circuit baseline did not: random sets given the same coefficient fitting failed at up to ~1,100 latents, but at 2,000–6,000 latents they reproduced the target under zero ablation (0.71–1.00) while still failing necessity. The cause is the encoder's sparsity: 6.5–37% of a target's live ReLU latents are nonzero at its anchor, against 0.3% expected under Top-K ($k = 128$ of 40,960), so under ReLU a random latent's fitted coefficient has a nonzero value to scale, and thousands of them can synthesise the target; under Top-K a random latent is almost always zero at the anchor. On dense dictionaries, necessity rather than faithfulness is the discriminating test, and the random baseline needs its own design.

# T Weighted Circuit Masking Design Ablations

Rejected or superseded designs for WCM, recorded because each closes a natural reviewer question.

## T.1 Loss Terms for Target Rank and Specificity

An *off-target* term charging every other site latent for rising above its unmodified and empty-circuit values cut lifts three- to sevenfold at no faithfulness cost, but addressed a property the method need not guarantee (Appendix J) and grew circuits when it also charged harmless lifts below the cut. A *Top-K hinge* requiring only that the target beat the $k$-th other latent restored faithfulness but, used alone, let circuits keep the target in the Top-K by lifting it together with its neighbourhood. A convex pull towards rank 1 inflated circuits by an order of magnitude at every weight tried, since it never switches off and dominates once the reproduction error is small. The *firing-margin* objective, fitting the target's lead over the Top-K cut, needs its loss normalised by the target's pre-activation scale rather than the margin's, which is near zero for targets at the cut. Rank-keep, which switches off once the target holds its natural rank, is the single term that kept the gains without these costs, and its effect is flat across a thirtyfold range of weights.

## T.2 Normalisation

The shared normaliser is target-scaled and bounded by construction because its predecessor, in an earlier two-method objective, was not: normalising each term by its own fully-closed-mask loss, which is unbounded far from the data, silently annihilated the zero-ablation term at depth (the two closed-mask losses differ there by up to eight orders of magnitude), reducing WCM[Z+C] training to the C term alone; the design rule is that a normaliser must never be able to annihilate its own term.

## T.3 Training under One Ablation Method

In pre-v1 calibration the A term roughly halves the smallest passing node set on layer-2 targets and raises the share of layer-9 targets with any passing circuit. Removing the zero-ablation term from training explodes zero-ablation faithfulness to 36–1,681 at depth while the trained ablation methods stay in band: coefficients trained without it calibrate against the mean ablations' background and overshoot by one to three orders of magnitude once that background is removed.

## T.4 Binarisation Sweep

The gates are optimised directly rather than sampled (Louizos et al., 2018; Li and Janson, 2024); the local minima that sampling avoids appear here as the equivalence class of near-optimal node sets (Appendix O). The straight-through estimator makes interior gate values forward-irrelevant, so the sparsity penalty drifts nodes toward just-above-threshold: the decision boundary becomes an *attractor*, and per-step node flips grow through training (218 per step at layer 10), a stopping-step lottery. Annealing instead yields the tightest zero-ablation faithfulness in the programme (0.002–0.10 deviation from unity) at slightly worse mean-ablation scores, another home-turf demonstration. Schedule compression does not help: a 200-step run matches the 400-step metrics at ~2× the per-step churn (churn is set by per-step pressure, not schedule shape); reach-then-hold leaves churn unchanged

and inflates size 21–48%; and truncating a 400-step run at 200 yields node sets 57–72% oversized with the gate still half-soft.